KON 313
Feedback Control Systems

Time Response

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Review

- After the engineer obtains a mathematical representation of a system, the system is **analyzed** for its response to some inputs to see if its characteristics yield the desired behavior.
- The output response of a control system consists of two parts:

  Transient response. which goes from the initial state to the final state.

  Steady-state response. the way the system output behaves as t approaches infinity.

$$c(t) = c_{tr}(t) + c_{ss}(t)$$

- Here, we analyze the transient response.
- Note that all systems having the same transfer function will exhibit the same output in response to the same input.

Poles, Zeros, and System Response

- Although many techniques, such as solving a diff. eqn. or taking the inverse Laplace transform, enable us to evaluate the output response of a control system, they are laborious and time-consuming.

- We prefer to use analysis and design techniques that yield results in a minimum of time.

- The concept of poles and zeros and their relationship to the time response of a system is a technique which simplifies the evaluation of a system's response.

Poles, Zeros, and System Response

Poles of a Transfer Function are

1. the values of the Laplace transform variable, s, that cause the transfer function to become infinite, or
2. any roots of the denominator of the transfer function that are common to roots of the numerator.

Zeros of a Transfer Function are

1. the values of the Laplace transform variable, s, that cause the transfer function to become zero, or
2. any roots of the numerator of the transfer function that are common to roots of the denominator.

Poles, Zeros, and System Response

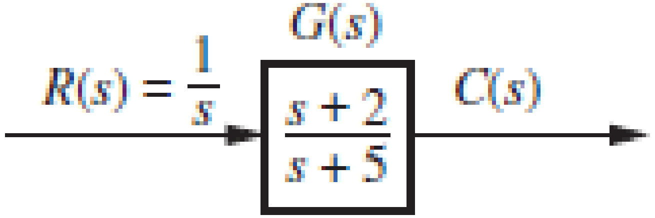
- **Order of a System.** The order refers to the order of
  - the equivalent differential equation representing the system, or
  - the order of the denominator of the transfer function after cancellation of common factors in the numerator, or
  - the number of simultaneous first-order equations required for the state-space representation.
- **Pole-Zero Plot of a System.** These values are plotted on the complex s-plane, using an × for each pole and a ○ for each zero.

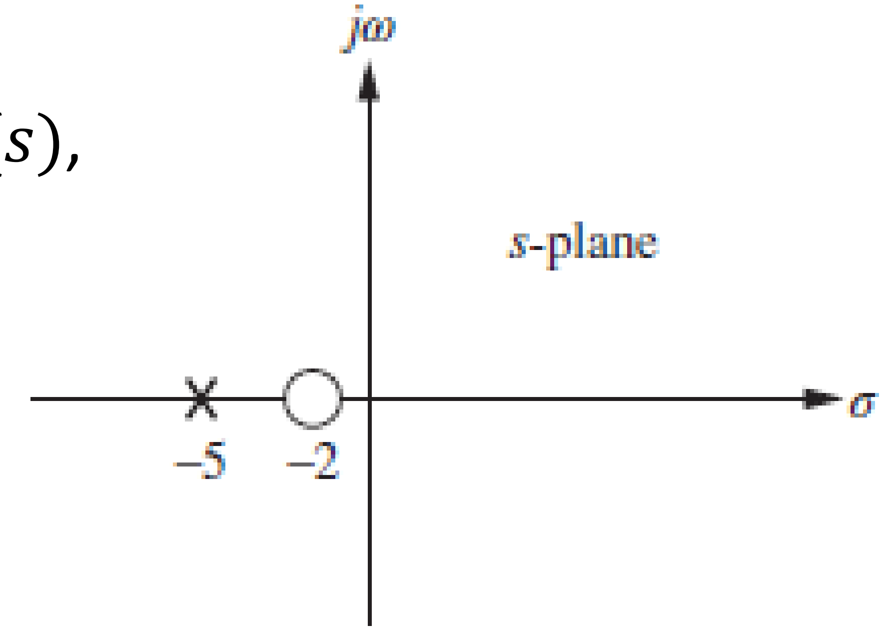
Poles, Zeros, and System Response

Ex. first order system. Given the transfer function $G(s)$,

a pole exists at -5, and

a zero exists at -2.





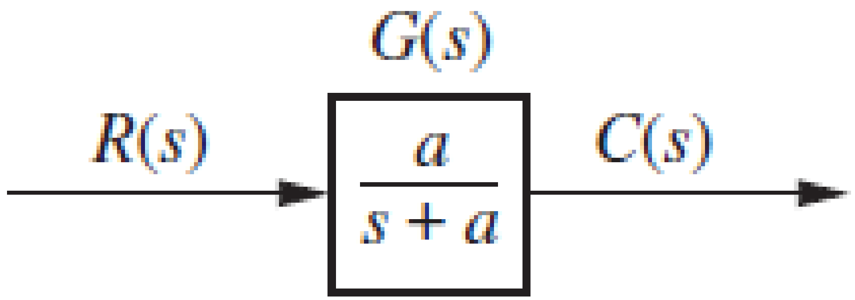
Consider the unit step response of the system:

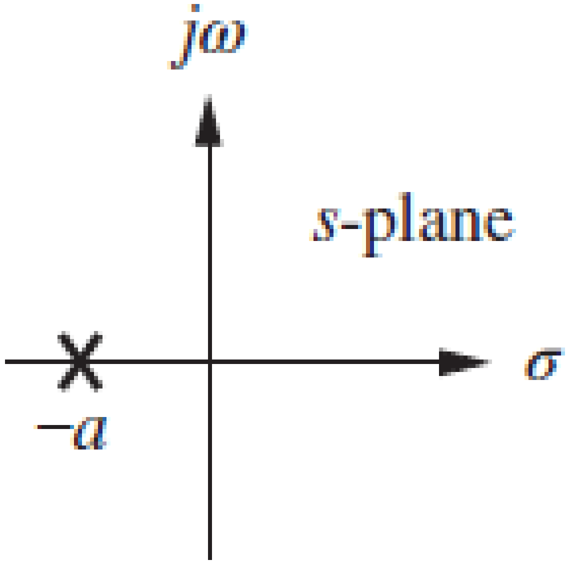
$$C(s) = \frac{(s+2)}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} = \frac{2/5}{s} + \frac{3/5}{s+5} \quad \Longrightarrow \quad c(t) = \left(\frac{2}{5} + \frac{3}{5}e^{-5t}\right)u(t)$$

1. A pole of the input function generates the form of the forced response,
2. A pole of the trans. function generates the form of the natural response,
3. The zeros and poles generate the amplitudes for both the forced and natural responses.

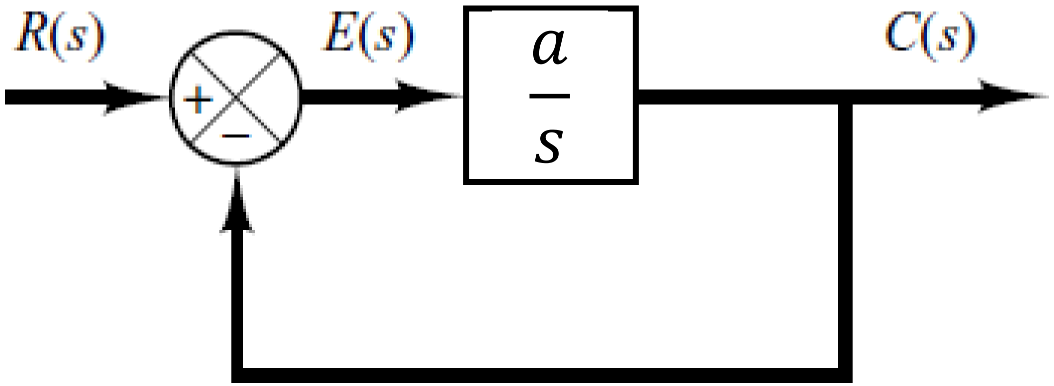
First-Order Systems

A first-order system without zeros





Closed-loop block diagram representation:



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First-Order Systems

Unit-Step Response. Given $R(s) = 1/s$, we obtain

$$C(s) = R(s)G(s) = \frac{a}{s(s+a)}$$

Taking the inverse Laplace transform

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$

where the input pole at the origin generated the forced response $c_f(t)$, and the system pole at $-a$, generated the natural response $c_n(t)$.

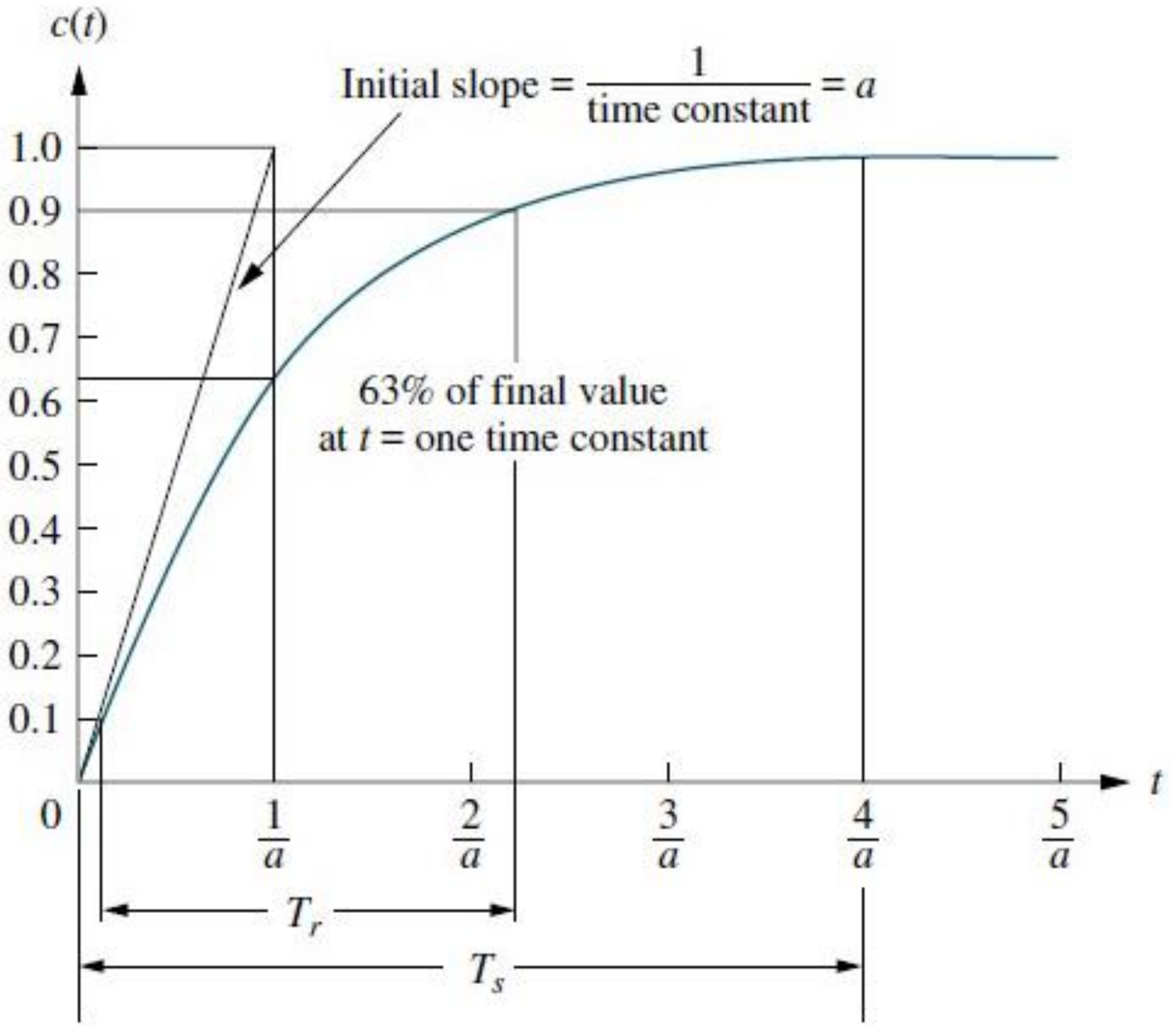
Thus, the **only parameter** needed to describe the transient response is a.

First-Order Systems

Unit-Step Response. Transient response performance specifications

- Time Constant
- Rise Time
- Settling Time

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$



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First-Order Systems

Unit-Step Response.

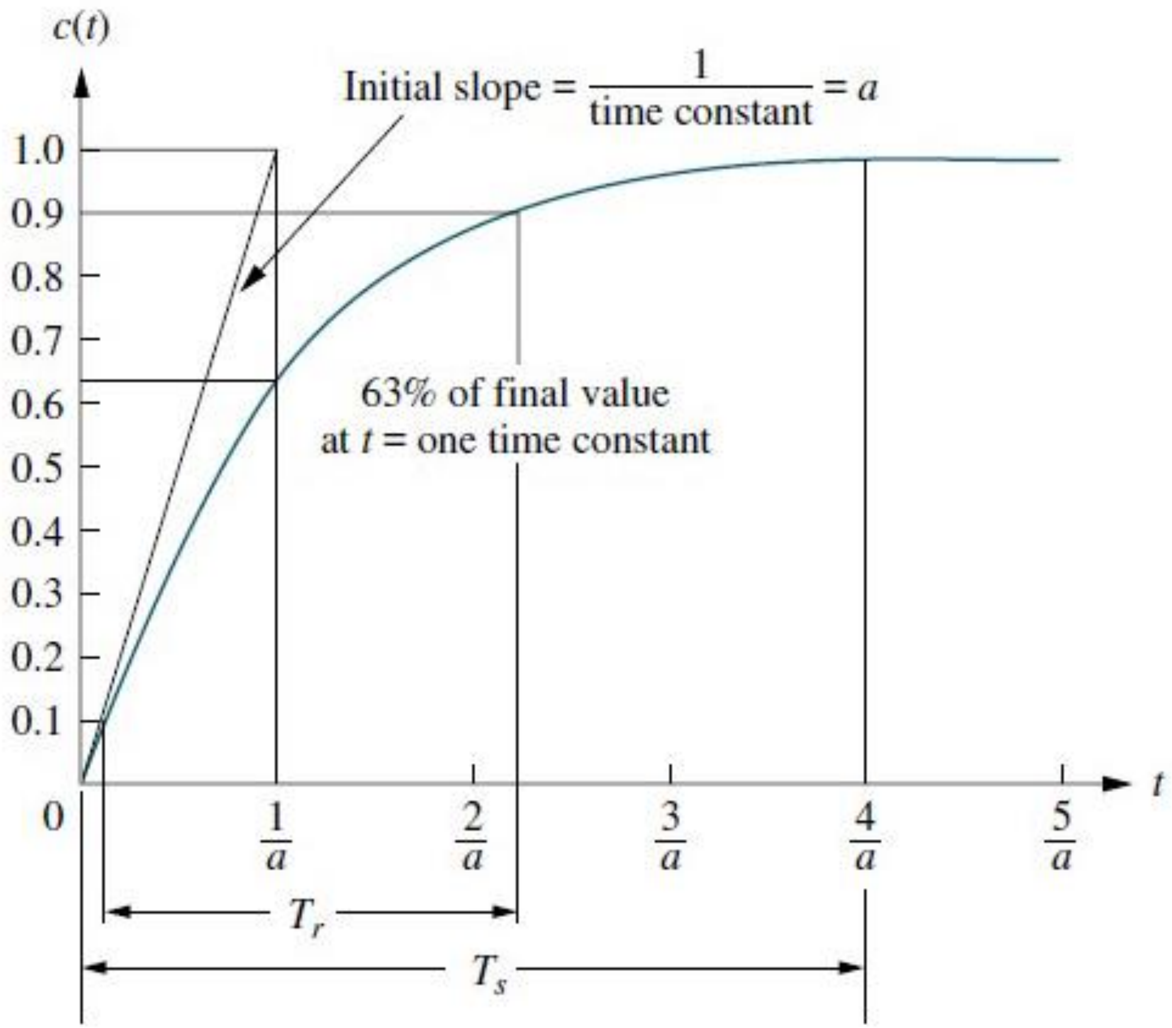
- *Time Constant.*

$\mathrm{T} = 1/a$ is the time constant of the response.

- The time constant is the time for e^{-at} to decay to 37% of its initial value.

$$e^{-at}\big|_{t=1/a} = e^{-1} = 0.37$$

- Alternately, the time constant is the time it takes for the step response to rise to 63% of its final value.



$$c(t)\big|_{t=1/a} = 1 - e^{-at}\big|_{t=1/a} = 1 - 0.37 = 0.63$$

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First-Order Systems

Unit-Step Response.

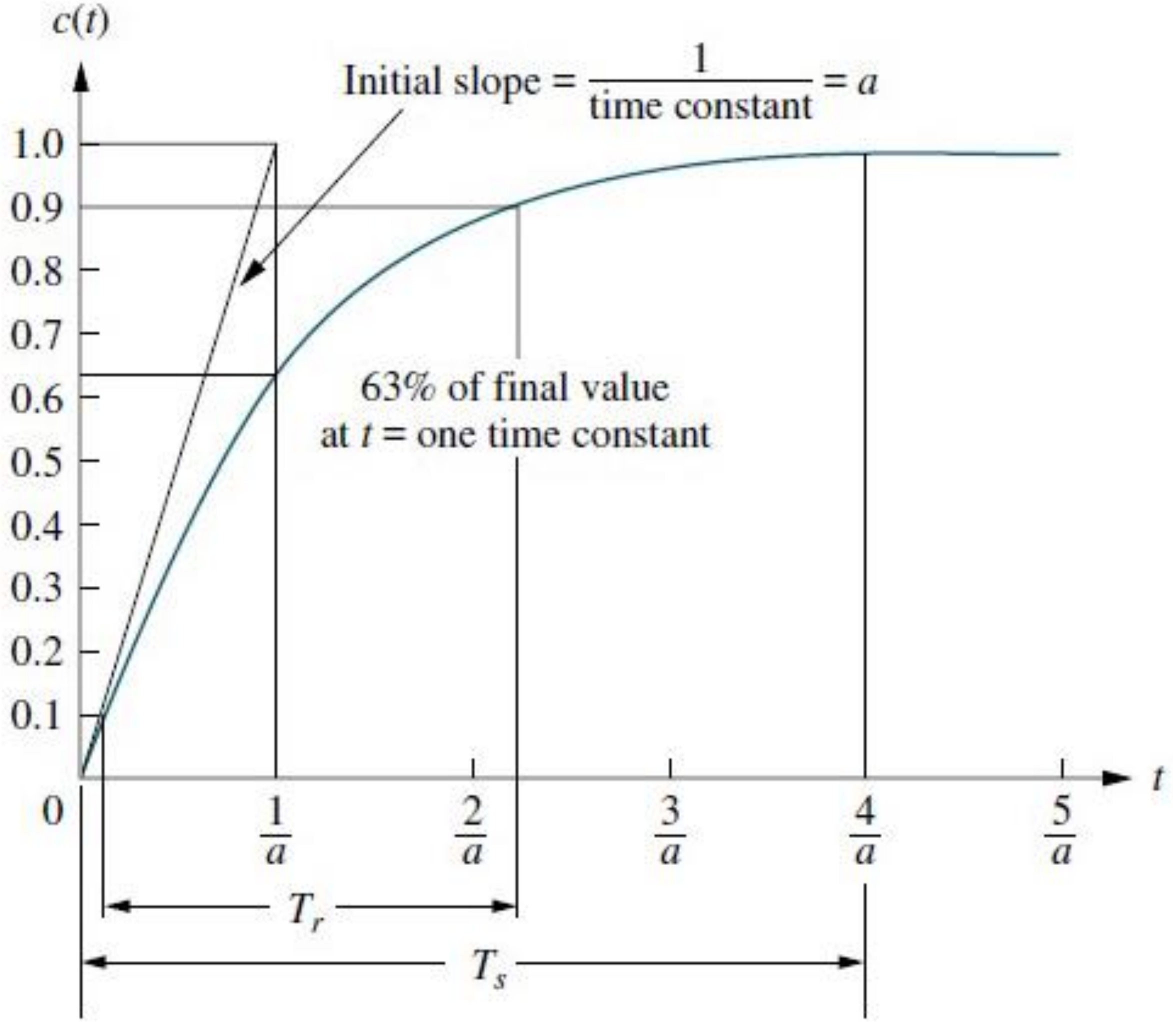
- *Time Constant*

- The parameter a has the units (1/sec), or frequency, and is called *exponential frequency*.

- The slope of the tangent line at $t = 0$ is a

$$\left.\frac{dc(t)}{dt}\right|_{t=0} = \left.ae^{-at}\right|_{t=0} = a$$

$\Longrightarrow$ a is the initial rate of change at $t = 0$.



$\Longrightarrow$ Time constant is related to the speed at which the system responds to a step input.

- The pole of the transfer function is located at $-a$,

  $\Longrightarrow$ the farther the pole from the imaginary axis, the faster the transient response.

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First-Order Systems

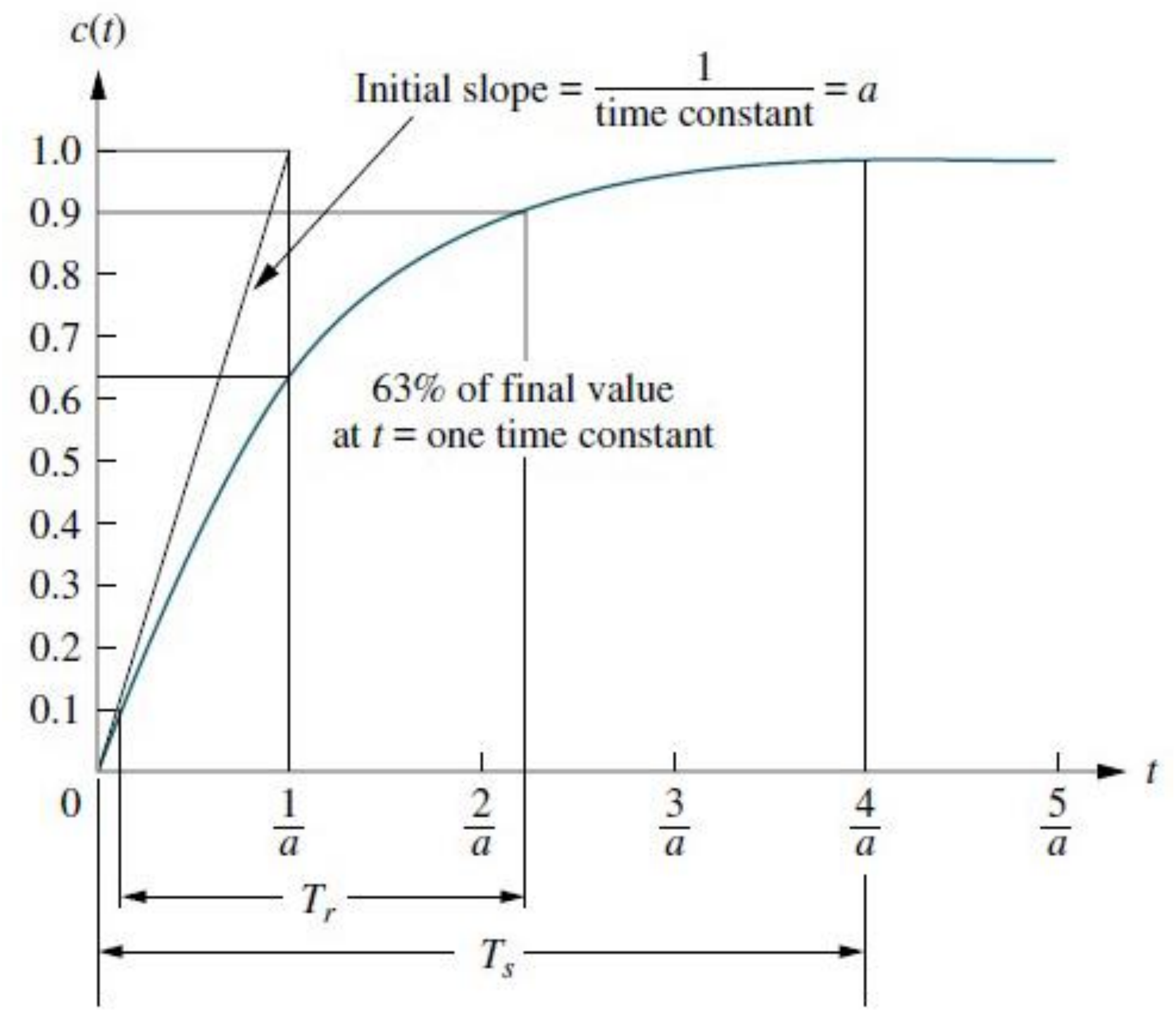
Unit-Step Response.

- *Rise Time.* the time for the step response to go from 0.1 to 0.9 of its final value. It is found by solving $c(t) = 1 - e^{-at}$ for the difference in time at $c(t) = 0.9$ and $c(t) = 0.1$.

$$T_r = \frac{2.31}{a} - \frac{0.11}{a} = \frac{2.2}{a} = 2.2T$$

- *Settling time.* the time for the response to reach, and stay within 2% of its final value. Letting $c(t) = 0.98$ and solving for t, we find

$$T_s = \frac{4}{a} = 4T$$



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First-Order Systems

First-Order Transfer Functions via Testing.

• Often it is not possible to obtain a real system's transfer function analytically.

• In this case, the system's **step response** can be practically used to represent the system even though the inner construction is not known.

• We can **measure** the time constant and the steady-state value from the **step** response, then the transfer function can be estimated. Consider a simple first order system,

$$G(s) = \frac{K}{s+a} \quad \Longrightarrow \quad C(s) = \frac{K}{s(s+a)} = \frac{K/a}{s} - \frac{K/a}{(s+a)}$$

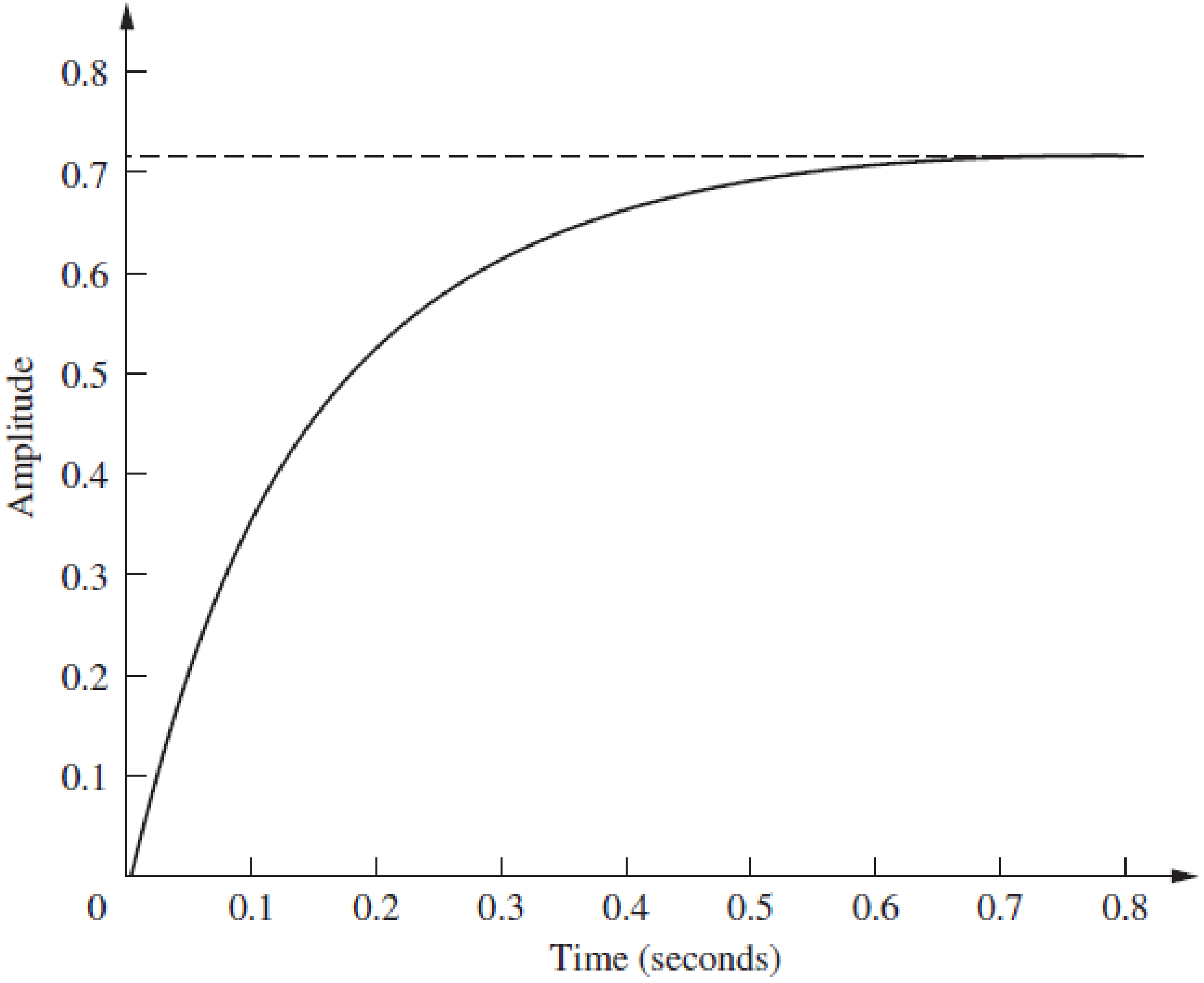
If we can identify K and a from laboratory testing, we can obtain the transfer function of the system.

Note that, the time response is $$c(t) = \frac{K}{a}(1 - e^{-at})$$

First-Order Systems

First-Order Transfer Functions via Testing.

Ex. Assume the unit step response for a system is recorded at the lab. Construct a transfer function for this system.



Sln. It seems it has a **first-order system** characteristics we have seen so far, such as no overshoot and nonzero initial slope.

- The **final value** is $\frac{K}{a} = 0.72$
- We can measure the **time constant**, that is the time for the amplitude to reach

63% of its final value. The curve reaches $0.63 \times 0.72 \cong 0.45$ at $T = 0.13$.

$$a = \frac{1}{T} = 7.7, \quad K = 0.72a = 5.54 \quad \Longrightarrow \quad G(s) = \frac{5.54}{s+7.7}$$

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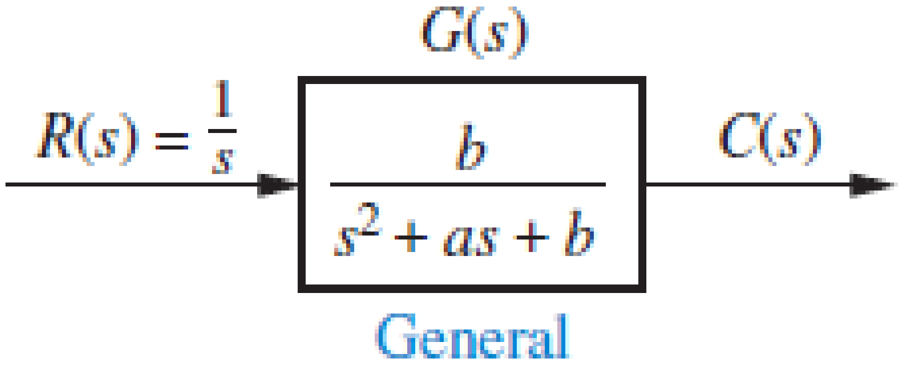
Second-Order Systems

A Second-order system without zeros

Compared to the simplicity of a first-order system, a second-order system exhibits a **wide range of responses**.

Whereas varying a first-order system's parameter simply changes the speed of the response, changes in the parameters of a second-order system can also change **the form of the response**.

We can use the **poles to determine the nature** of the response without going through the procedure of a partial-fraction expansion followed by the inverse Laplace transform.



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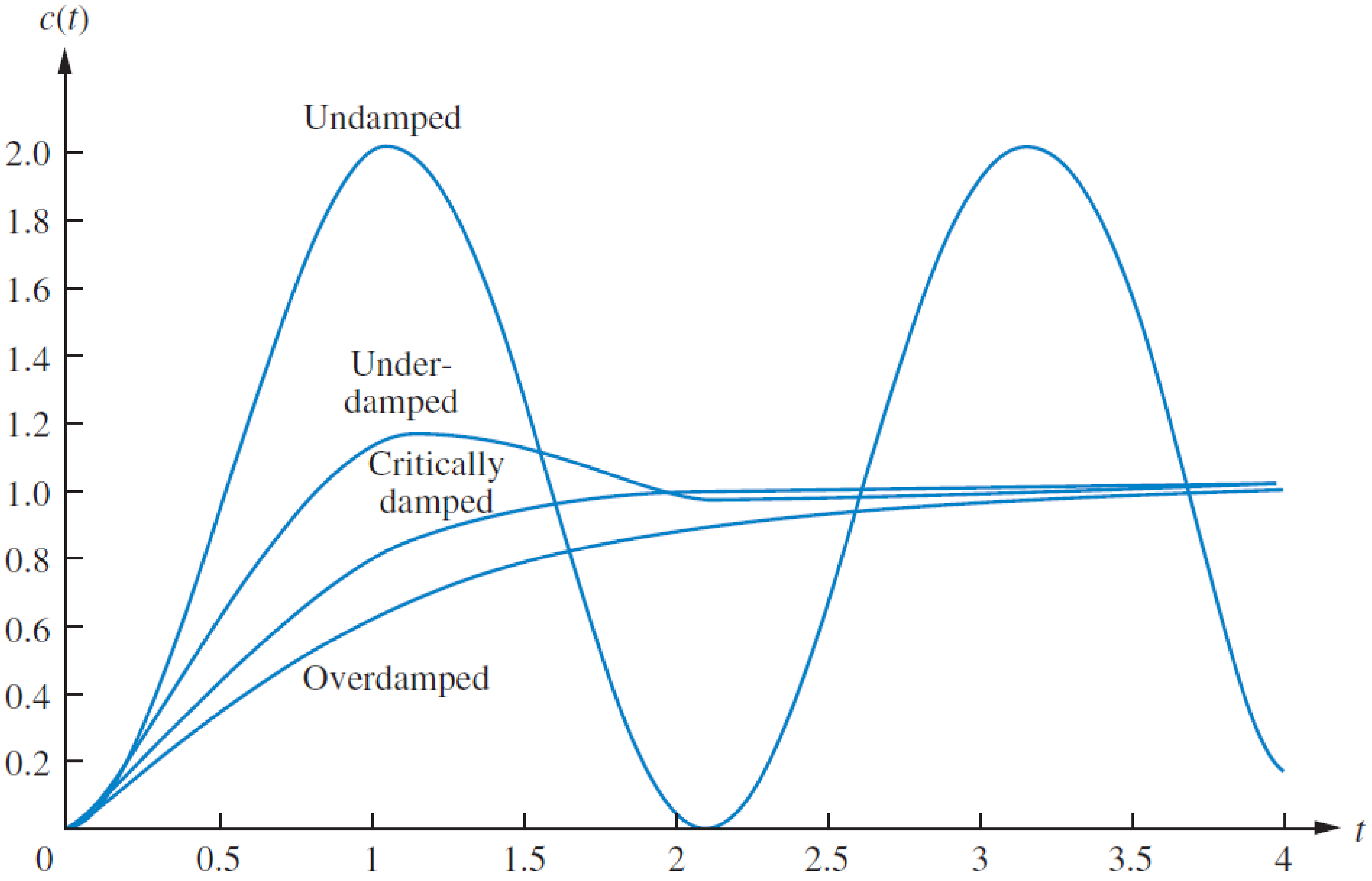
Second-Order Systems

Unit-Step Response. Four different cases might happen.



Second-Order Systems

Unit-Step Response. Overdamped Response

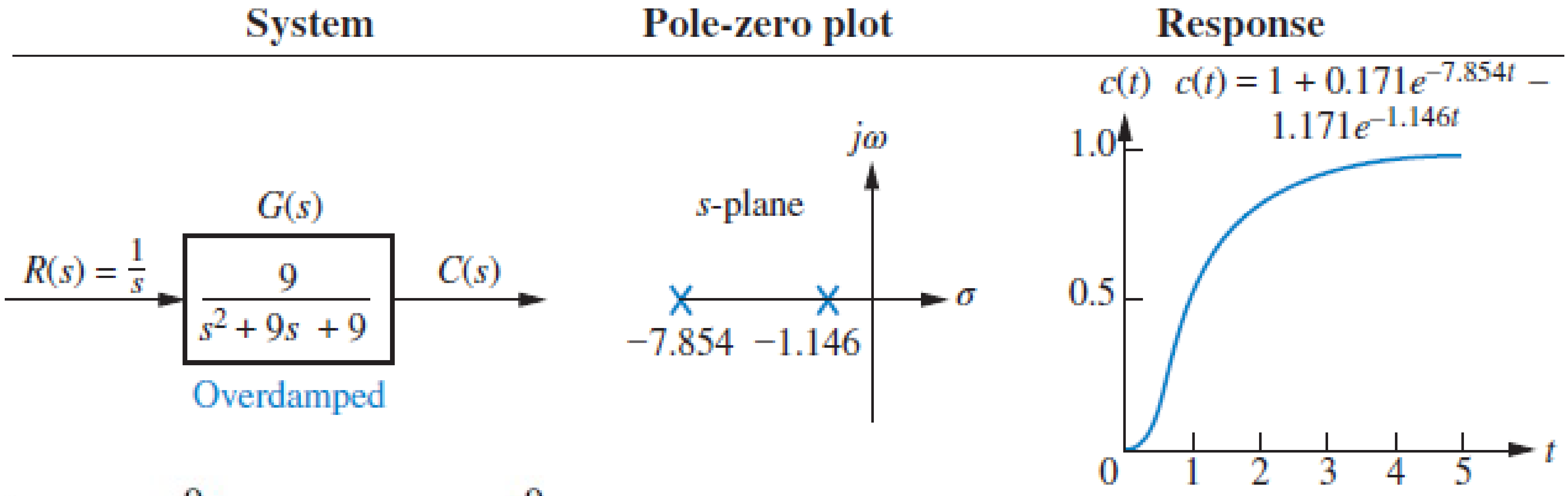
Poles: Two real at $-\sigma_1, \ -\sigma_2$

Natural response: Two exponentials with time constants equal to the reciprocal of the pole locations, or

$$c(t) = K_1 e^{-\sigma_1 t} + K_2 e^{-\sigma_2 t}$$

Second-Order Systems

Unit-Step Response. Ex. Overdamped



$$C(s) = \frac{9}{s(s^2+9s+9)} = \frac{9}{s(s+7.854)(s+1.146)} \Longrightarrow c(t) = K_1 + K_2 e^{-7.854t} + K_3 e^{-1.146t}$$

- The input pole at the origin generates the constant forced response;
- Each of the two system poles on the real axis generates an exponential natural response whose *exponential frequency* is equal to the **pole location**.

Second-Order Systems

Unit-Step Response. Critically Damped Response

Poles: Two real at $-\sigma_1$

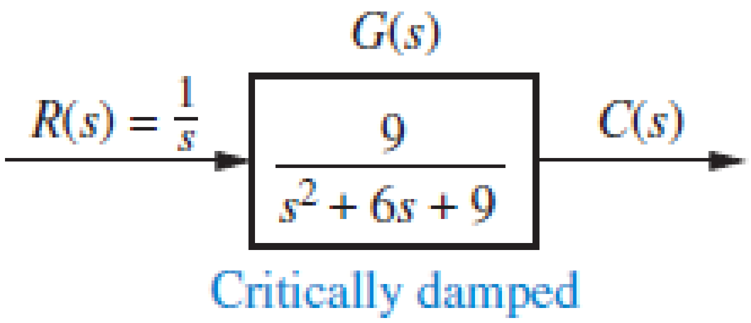
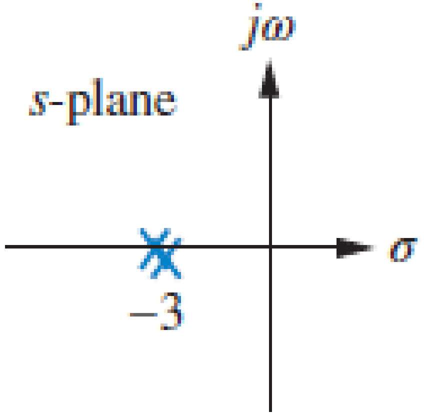
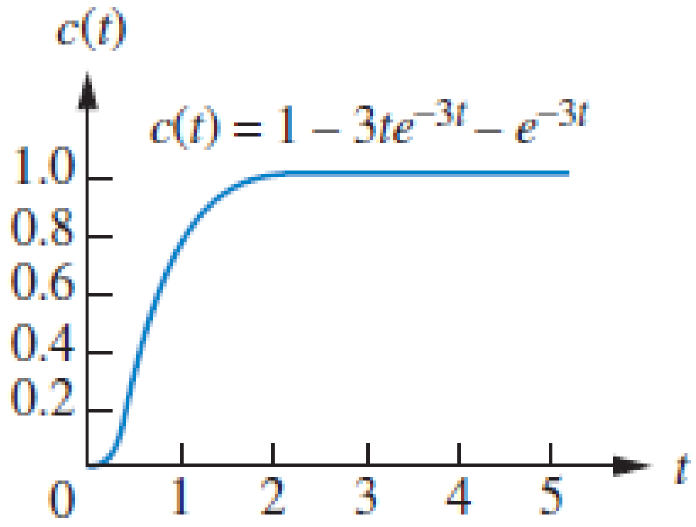
Natural response: One term is an exponential whose time constant is equal to the reciprocal of the pole location. Another term is the product of time, t, and an exponential with time constant equal to the reciprocal of the pole location, or

$$c(t) = K_1 e^{-\sigma_1 t} + K_2 t e^{-\sigma_1 t}$$

Notice that the critically damped case is the division between the overdamped cases and the underdamped cases and is the **fastest response without overshoot**.

Second-Order Systems

Unit-Step Response. Ex. Critically Damped

System	Pole-zero plot	Response
		

$$C(s) = \frac{9}{s(s^2+6s+9)} = \frac{9}{s(s+3)^2} \quad \Longrightarrow \quad c(t) = K_1 + K_2 e^{-3t} + K_3 t e^{-3t}$$

- The input pole at the origin generates the constant forced response, and
- The two poles on the real axis at −3 generate a natural response consisting of an exponential and an exponential multiplied by time.

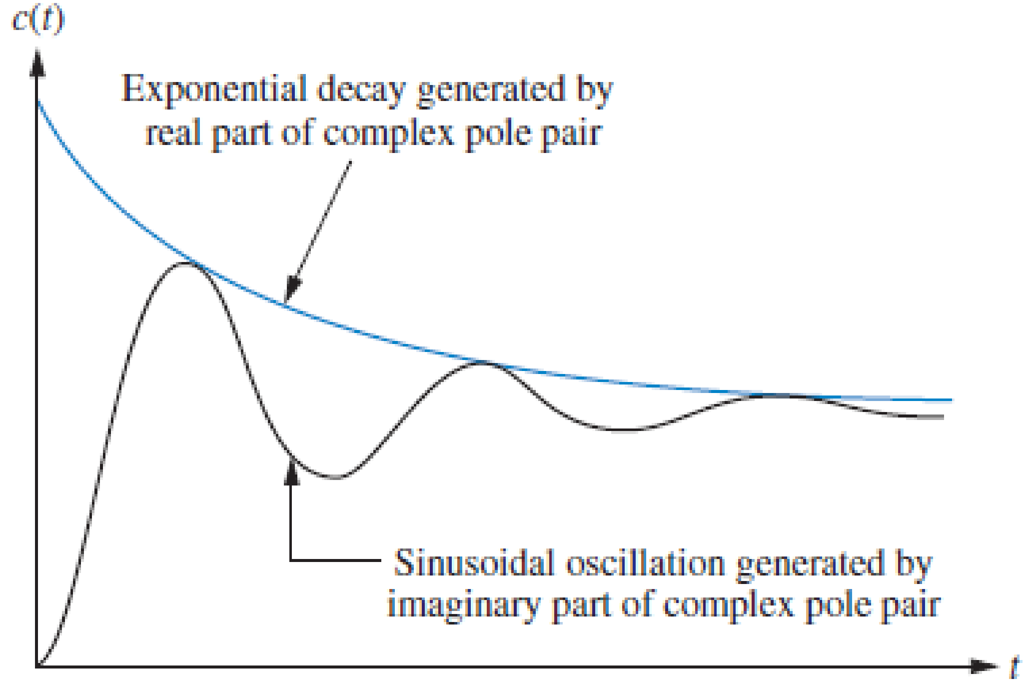
Second-Order Systems

Unit-Step Response. Underdamped Response

Poles: Two complex at $-\sigma_d \pm j\omega_d$

Natural response: Damped sinusoid with an exponential envelope whose time constant is equal to the reciprocal of the pole's real part. The radian frequency of the sinusoid, the damped frequency of oscillation, is equal to the imaginary part of the poles, or

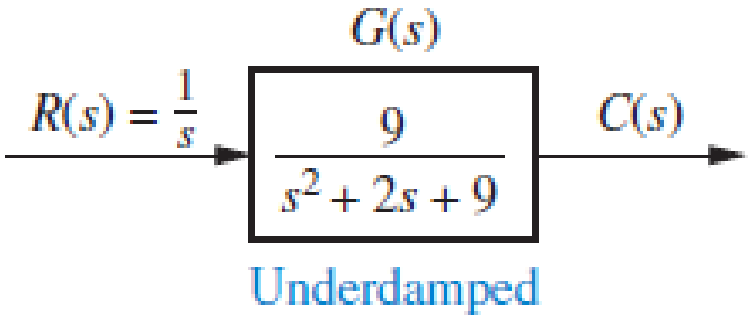
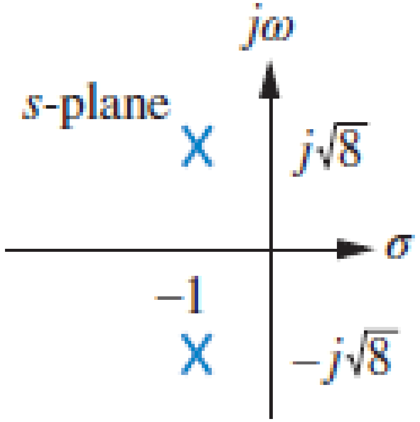
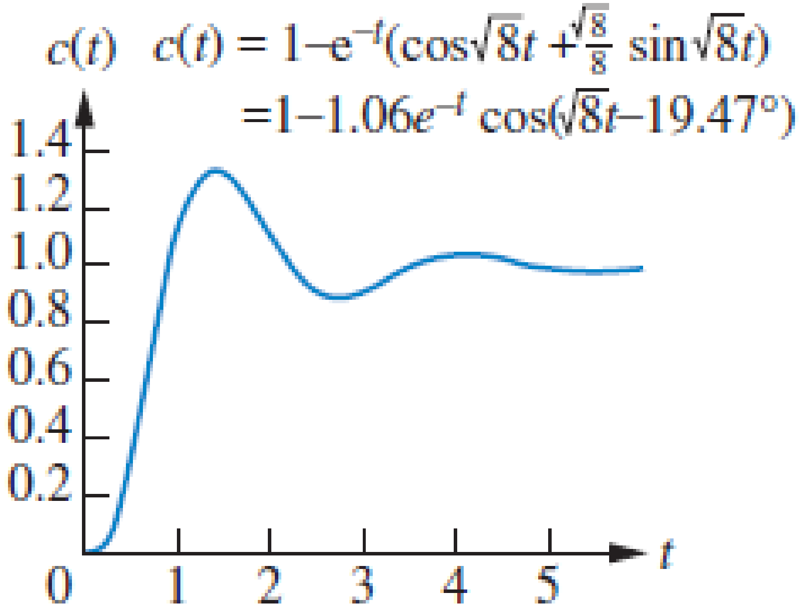
$$c(t) = Ae^{-\sigma_d t}\cos(\omega_d t - \phi)$$



- The real part of the pole matches the *exponential decay frequency* of the sinusoid's amplitude,
- The imaginary part of the pole matches the frequency of the sinusoidal oscillation (*damped frequency of oscillation*, ω_d).

Second-Order Systems

Unit-Step Response. Ex. Underdamped

System	Pole-zero plot	Response
		

$$C(s) = \frac{9}{s(s^2+2s+9)}$$

$$c(t) = K_1 + e^{-t}\left(K_2\cos\left(\sqrt{8}t\right) + K_3\sin\left(\sqrt{8}t\right)\right) = K_1 + K_4 e^{-t}\cos(\sqrt{8}t - \phi)$$

where $\phi = \tan^{-1}K_3/K_2$, $K_4 = \sqrt{K_2^2 + K_3^2}$, $\mathbf{c(t) = 1 - 1.06e^{-t}\cos(\sqrt{8}t - 19.47°)}$

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Second-Order Systems

Unit-Step Response. Undamped Response

Poles: Two imaginary at $\pm j\omega_1$

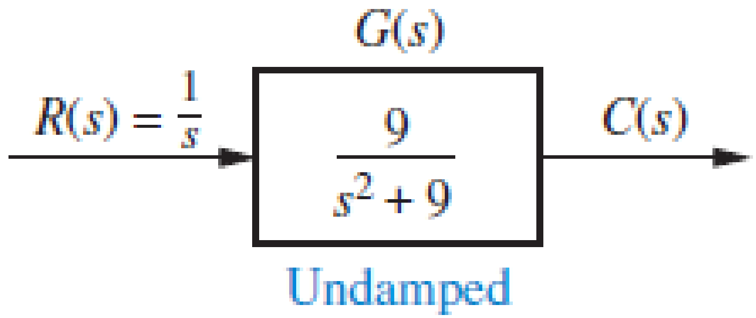
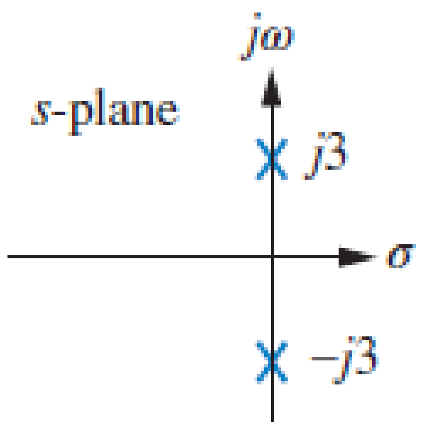
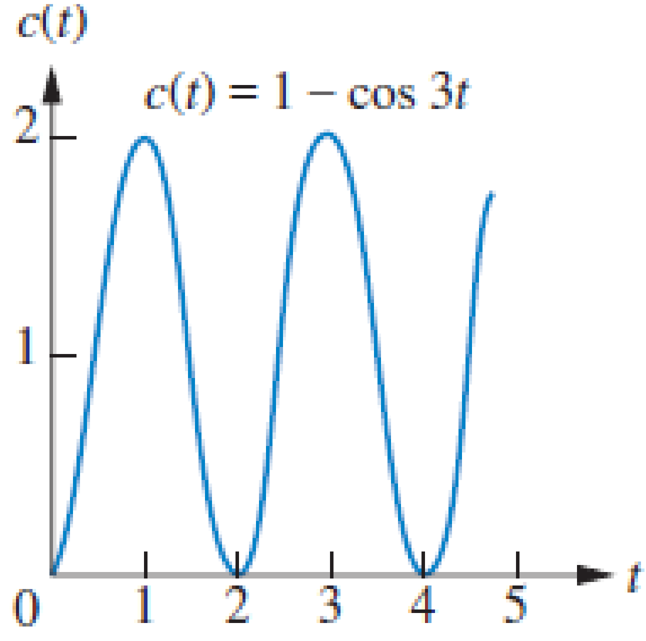
Natural response: Undamped sinusoid with radian frequency equal to the imaginary part of the poles, or

$$c(t) = \mathrm{A}\cos(\omega_1 t - \phi)$$

Second-Order Systems

Unit-Step Response. Ex. Undamped

- The input pole at the origin generates the constant *forced* response, and

System	Pole-zero plot	Response
		

$$C(s) = \frac{9}{s(s^2+9)} \quad \Longrightarrow \quad c(t) = K_1 + K_4 \cos(3t - \phi).$$

- The two system poles on the imaginary axis at $\pm j3$ generate a sinusoidal *natural* response whose frequency is equal to the location of the imaginary poles.
- Note that the absence of a real part in the pole pair corresponds to an exponential that does not decay. $e^{-0t} = 1$

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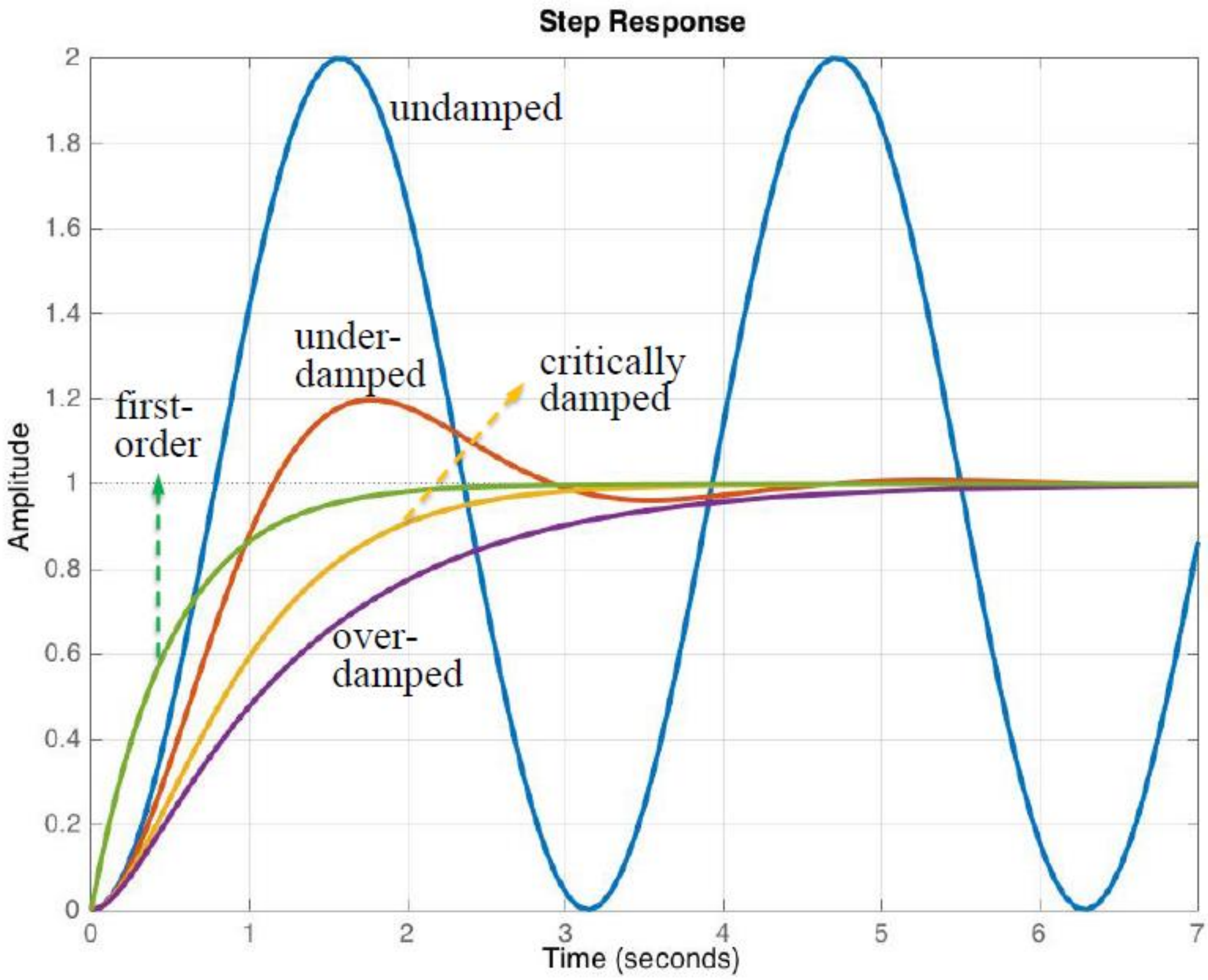
Second-Order Systems
Unit-Step Response – Damping Cases.

Overdamped Responses	Poles:	Two real at σ_1 and σ_2
	Natural Response:	$c(t) = K_1 e^{-\sigma_1 t} + K_2 e^{-\sigma_2 t}$
Underdamped Responses	Poles:	$-\sigma_d \mp j\omega_d$
	Natural Response:	$c(t) = Ae^{-\sigma_d t}\cos(\omega_d t - \phi)$
Undamped Responses	Poles:	$\mp j\omega_1$
	Natural Response:	$c(t) = A\cos(\omega_1 t - \phi)$
Critically Damped Responses	Poles:	Two real at $-\sigma_1$
	Natural Response:	$c(t) = K_1 e^{-\sigma_1 t} + K_2 t e^{-\sigma_1 t}$

Second-Order Systems

Unit-Step Response of 1s Order and 2nd–Order Systems Damping Cases.



Second-Order Systems

The General Second-Order Systems

Two quantities are used to describe the characteristics of the second-order transient response (just as time constants describe the first-order system),

- The *natural frequency,* ω_n, of a 2nd-order system is the **frequency** of oscillation of the system **without damping**.
- The *damping ratio,* ζ, compares the exponential decay frequency of the envelope to the natural frequency.

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/second)}} = \frac{1}{2\pi}\,\frac{\text{Natural period (seconds)}}{\text{Exponential time constant}}$$

The definition is derived from the need to quantitatively describe the **damped oscillation** of **underdamped** systems **regardless of the time scale**.

Second-Order Systems

The General Second-Order Systems

Consider the general system

$$G(s) = \frac{b}{s^2 + as + b}$$

Having the **purely imaginary poles** ($a = 0$), the response would be an **undamped** sinusoid. The **natural frequency**, ω_n, is the frequency of oscillation of this system.

$$G(s) = \frac{b}{s^2 + b} \quad \Longrightarrow \quad \omega_n = \sqrt{b} \quad \Longrightarrow \quad b = \omega_n^2$$

Assuming an **underdamped** system, the complex poles have a **real part**, σ, equal to $-a/2$. The magnitude of this value is then the **exponential decay frequency**

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/second)}} = \frac{|\sigma|}{\omega_n} = \frac{a/2}{\omega_n} \quad \Longrightarrow \quad a = 2\zeta\omega_n$$

Second-Order Systems

The General Second-Order Systems

The *standard form* of the second-order system is thus:

$$\boxed{G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

Solving for the poles of the general transfer function yields:

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

The nature of the response obtained is related to the value of ζ. Variations of damping ratio **alone** yield the **complete range** of overdamped, critically damped, underdamped, and undamped responses.

Second-Order Systems

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

The General Second-Order Systems – Unit-step Response

- Undamped ($\zeta = 0$)

$$s_{1,2} = \pm j\omega_n \quad \xRightarrow{R(s)=\frac{1}{s}} \quad c(t) = 1 - \cos(\omega_n t)$$

- Under damped ($0 < \zeta < 1$)

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\sigma \pm j\omega_d$$

- Critically damped ($\zeta = 1$)

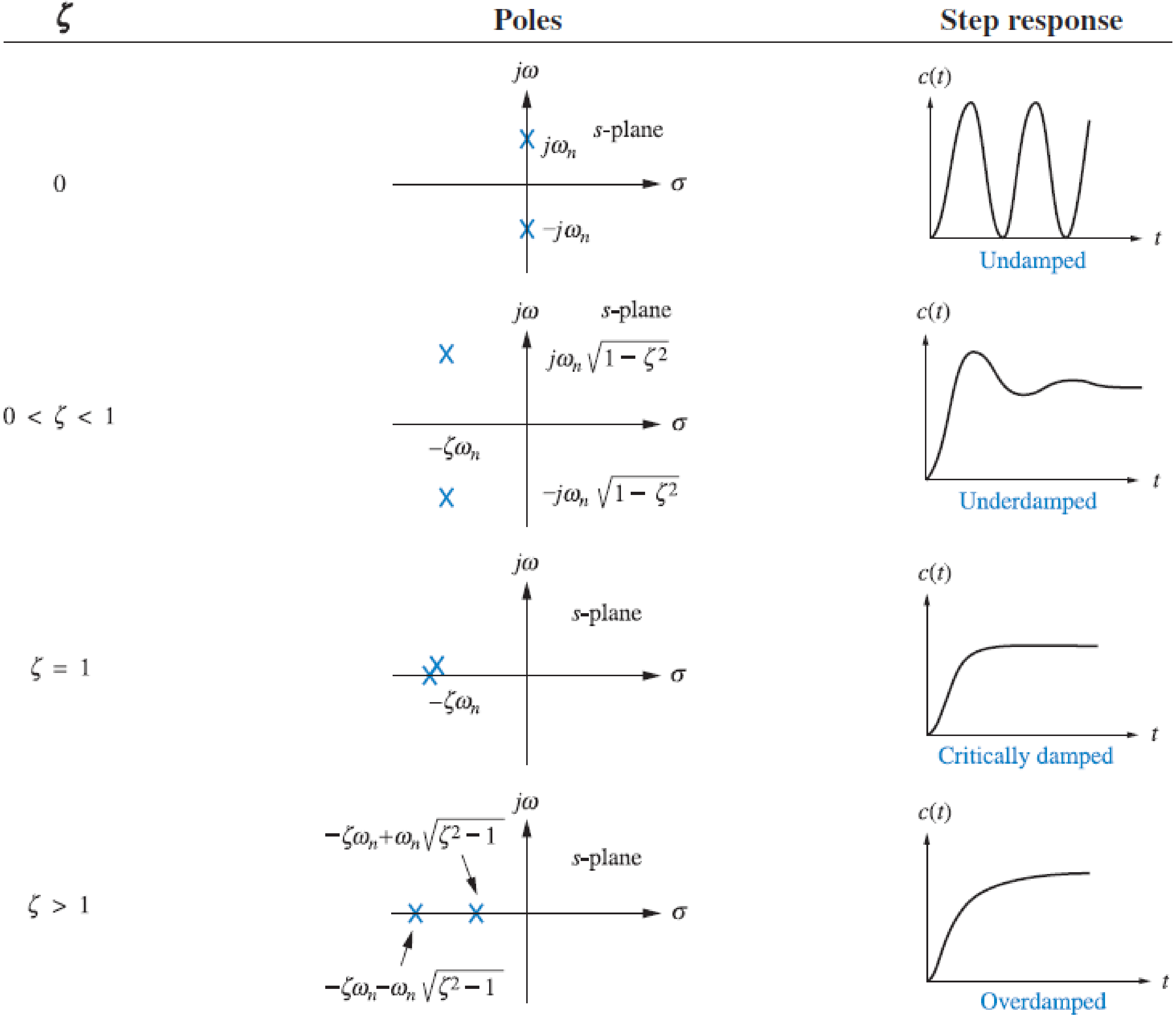
$$s_{1,2} = -\zeta\omega_n \quad \xRightarrow{R(s)=\frac{1}{s}} \quad c(t) = 1 - e^{-\omega_n t}(1 + \omega_n t)$$

- Overdamped ($\zeta > 1$)

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad \xRightarrow{R(s)=\frac{1}{s}} \quad c(t) = 1 + \frac{\omega_n}{2\sqrt{\zeta^2 - 1}}\left(\frac{e^{-s_1 t}}{s_1} - \frac{e^{-s_2 t}}{s_2}\right)$$

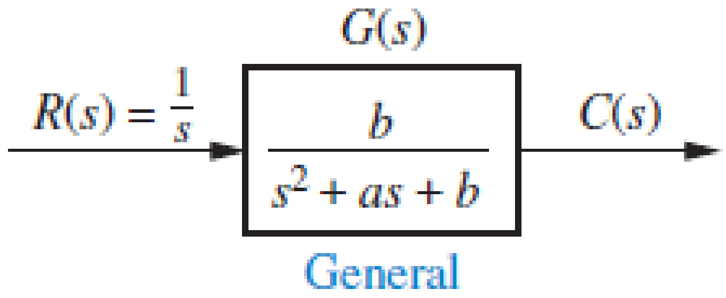
Second-Order Systems

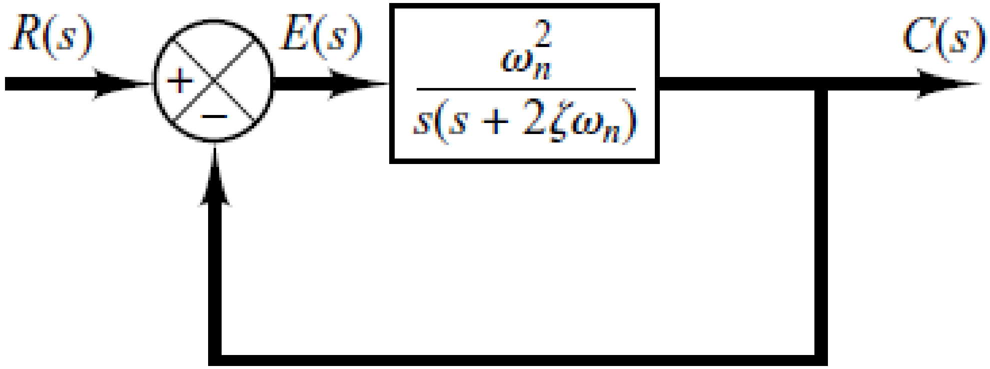
The General Second-Order Systems – Unit-Step Response



Second-Order Systems

Block diagram of a second order system





Q5-1

For a system with transfer function as $G(s) = \frac{200}{s+100}$, find the time value at which the step response reaches and stays within 2% of its final value.

a) 0.01

b) 0.02

c) 0.04

d) 0.08

e) None of the above

Answer: c

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Underdamped 2nd Order systems

- The underdamped second-order system, a **common model for physical problems**, displays unique behavior that must be itemized; a detailed description of the underdamped response is necessary for both **analysis** and **design**.

- We have already generalized the second-order transfer function in terms of ζ and ω_n as

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where it is assumed that $\zeta < 1$ that gives the underdamped system.

Underdamped 2nd Order systems

Unit-Step Response. Substituting $R(s) = 1/s$, expanding by partial fractions, and taking the inverse Laplace transform, we obtain

$$C(s) = R(s)G(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \dfrac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$\boxed{\begin{aligned} c(t) &= 1 - e^{-\zeta\omega_n t}\left(\cos\omega_n\sqrt{1-\zeta^2}\,t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_n\sqrt{1-\zeta^2}\,t\right) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\cos(\omega_n\sqrt{1-\zeta^2}\,t - \phi) \end{aligned}}$$

where $\phi = \tan^{-1}(\zeta/\sqrt{1-\zeta^2})$

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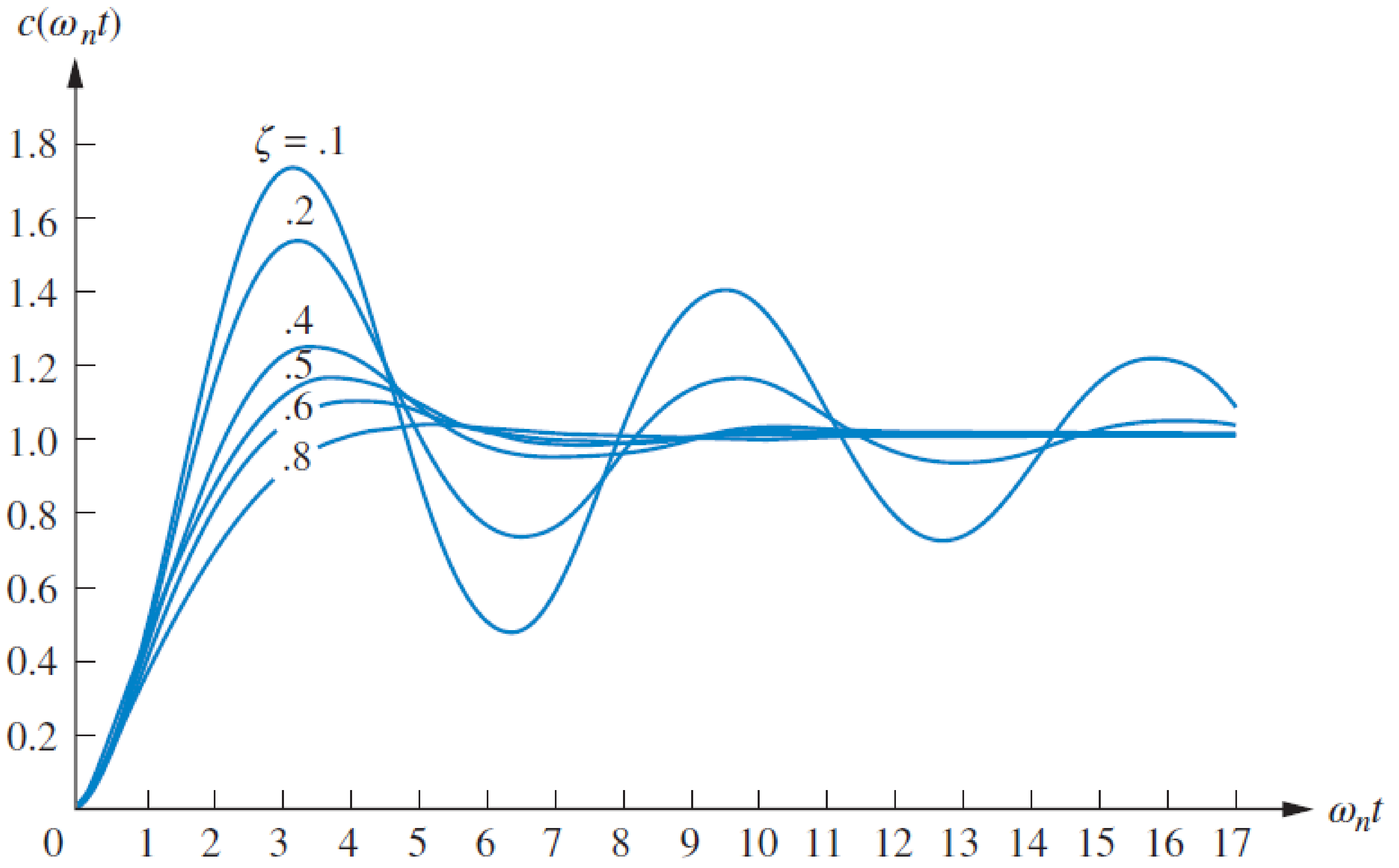
Underdamped 2nd Order systems

Unit-Step Response. The plot is shown for various values of ζ, plotted along a time axis **normalized** to the natural frequency, ω_n.

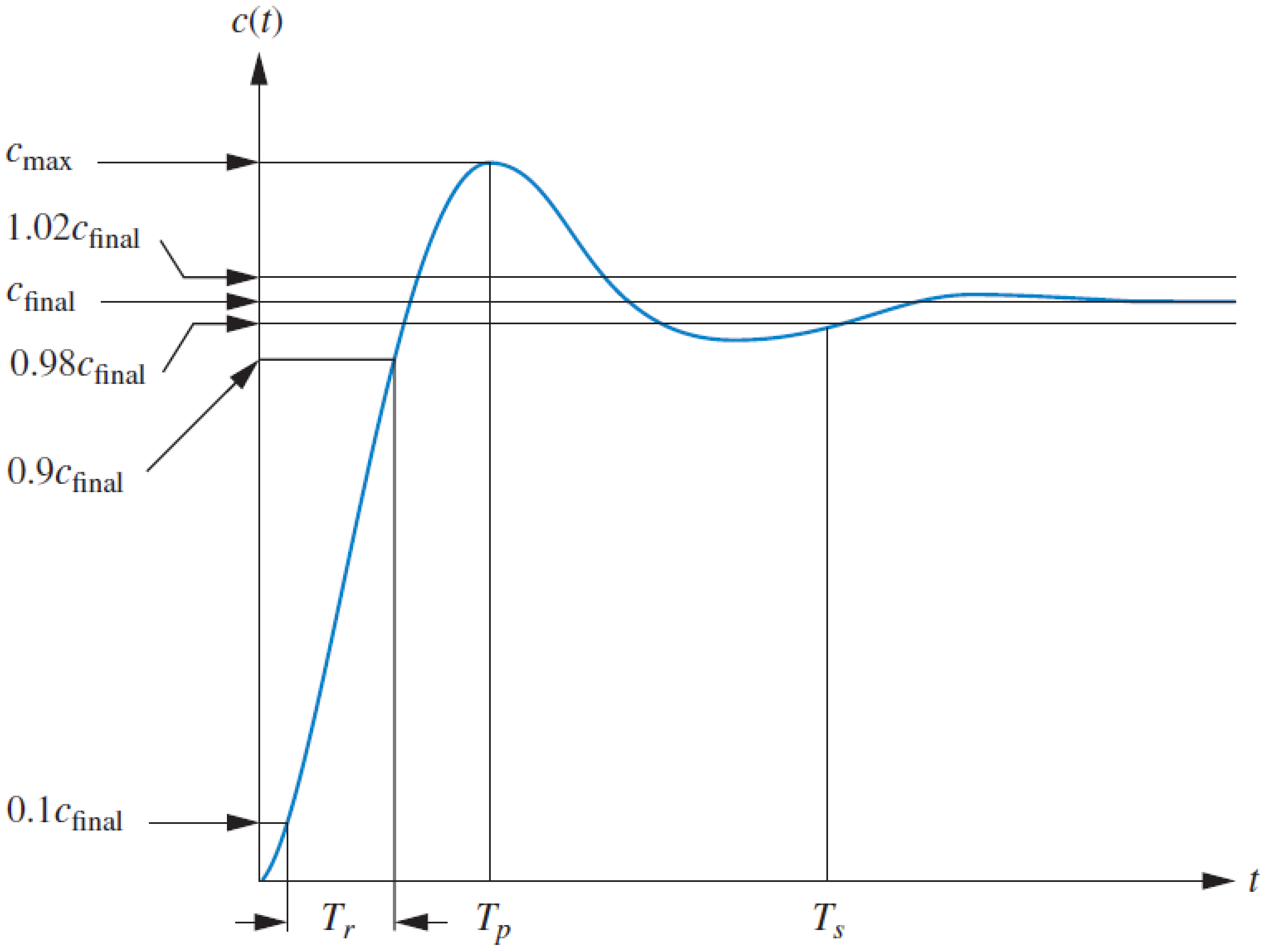
- The value of ζ affects the **type** of the response obtained: The **lower** the value of ζ, the **more oscillatory** the response.
- The ω_n is a **time-axis scale factor and does not affect the nature of** the response other than to scale it in time**.**



Underdamped 2nd Order systems

Unit-Step Response. Transient response performance specifications

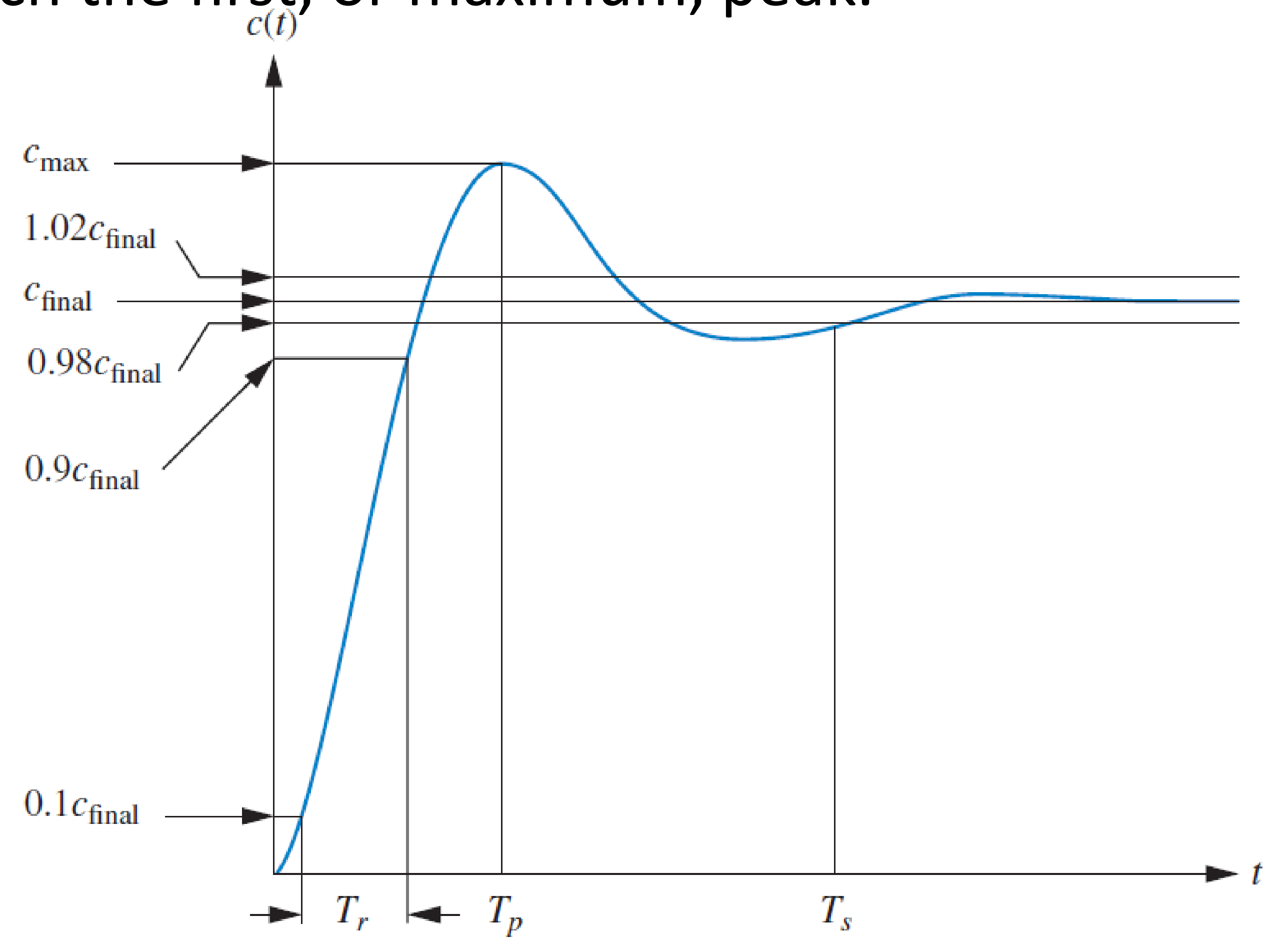
- Rise Time
- Peak Time
- Percent Overshoot
- Settling Time



Underdamped 2nd Order systems

Unit-Step Response.

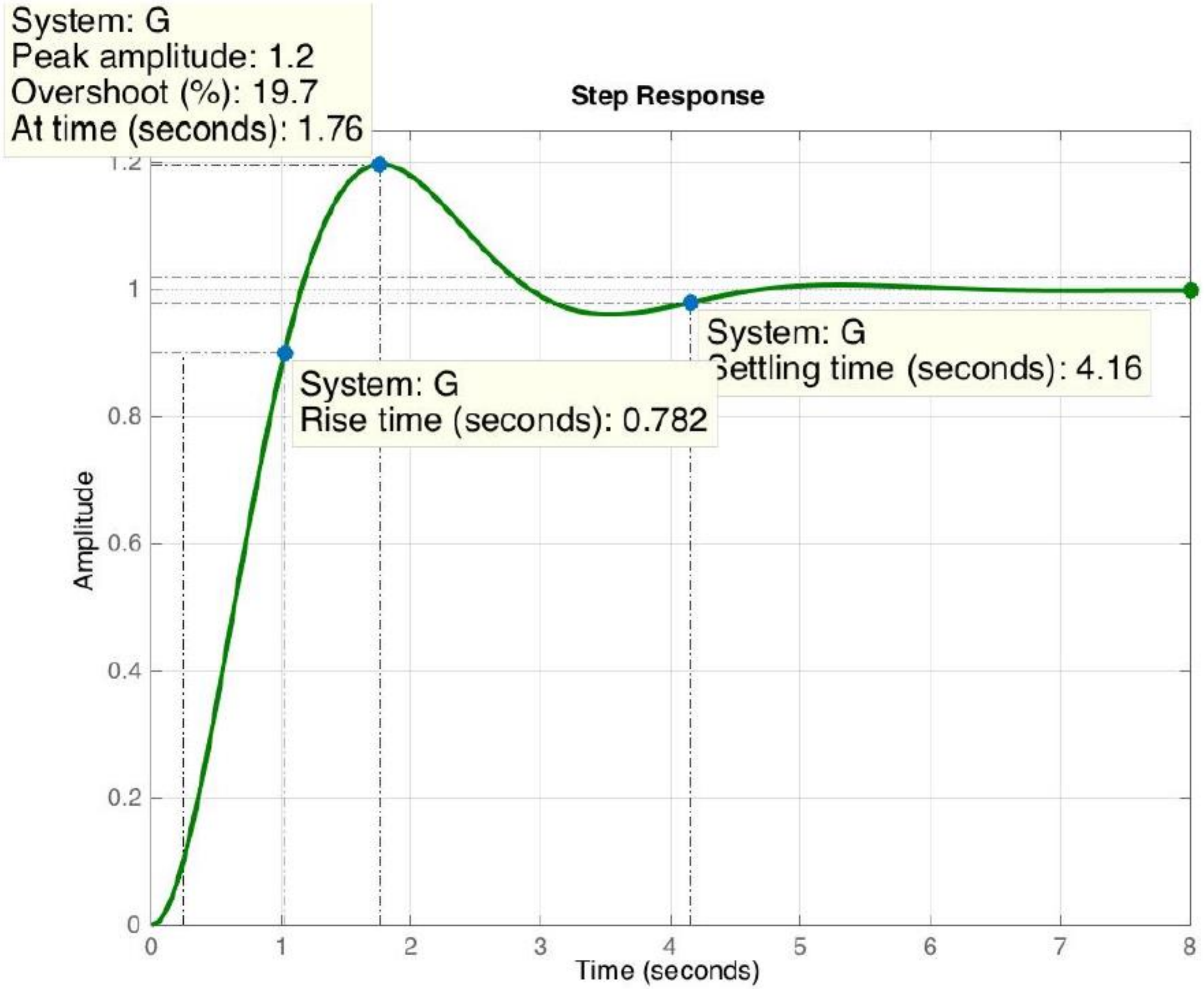
- *Rise Time,* T_r. The time required for the response to go from 0.1 of the final value to 0.9 of the final value.
- *Peak Time,* T_p. The time required to reach the first, or maximum, peak.
- *Percent Overshoot,* $\%OS$. The amount that the waveform overshoots the steady state, or final, value at the peak time, expressed as a percentage of the steady-state value.
- *Settling Time,* T_s. The time required for the transient's damped oscillations to reach and stay within ±2% of the steady-state value.



Underdamped 2nd Order systems

Unit-Step Response.

Ex.



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Underdamped 2nd Order systems

Unit-Step Response. The transient response specifications as function of ζ and ω_n.

Peak Time, T_p. is found by differentiating $c(t)$ and finding the first zero crossing after $t = 0$. By differentiating in the frequency domain, assuming zero initial conditions

$$\mathscr{L}[\dot{c}(t)] = sC(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Completing squares in the denominator,

$$\mathscr{L}[\dot{c}(t)] = \frac{\omega_n^2}{(s+\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$= \frac{\dfrac{\omega_n}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s+\zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$\dot{c}(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2}\, t$$

Setting the derivative equal to zero

$$\omega_n\sqrt{1-\zeta^2}\,t = n\pi \quad \text{or} \quad t = \frac{n\pi}{\omega_n\sqrt{1-\zeta^2}}$$

Each value of n yields the time for local maxima or minima. Letting $n = 0$ yields $t = 0$, the first point on the curve that has zero slope. The **first peak**, which occurs at the peak time, T_p, is found by letting $n = 1$.

$$\boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}$$

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Underdamped 2nd Order systems

Unit-Step Response. The transient response specifications as function of ζ and ω_n.

The percent overshoot, $\%OS$, is given by

$$\%OS = \frac{c_{\max} - c_{\text{final}}}{c_{\text{final}}} \times 100$$

the term c_{max} is found by evaluating $c(t)$ at the peak time, $c(T_p)$

$$\boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}$$

$$c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\cos(\omega_n\sqrt{1-\zeta^2}t - \phi)$$

$$\Longrightarrow \quad c_{\max} = c(T_p) = 1 - e^{-(\zeta\pi/\sqrt{1-\zeta^2})}\left(\cos\pi + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\pi\right) = 1 + e^{-(\zeta\pi/\sqrt{1-\zeta^2})}$$

For the unit step input, we know $c_{final} = 1$. Substituting these results, we get

$$\boxed{\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100}$$

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Underdamped 2nd Order systems

Unit-Step Response. The transient response specifications as function of ζ and ω_n.

The percent overshoot, $\%OS$,.

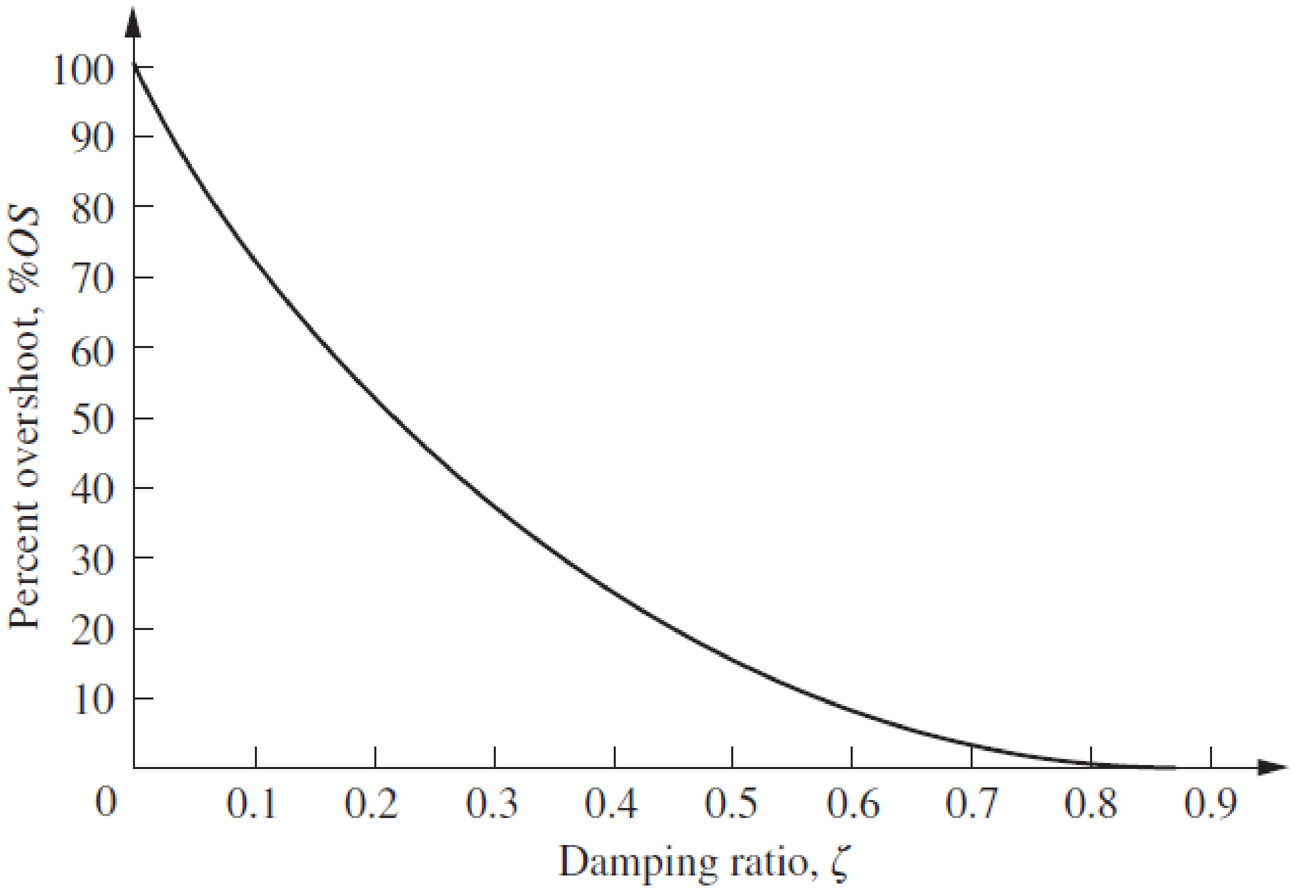
- Notice that the % overshoot is a function **only** of the damping ratio,

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

It allows us to find $\%OS$ for a given ζ,

- The inverse of the equation also allows us to solve for ζ for given $\%OS$.

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$



Underdamped 2nd Order systems

Unit-Step Response. The transient response specifications as function of ζ and ω_n.

Settling time, T_s. Using our definition, it is the time it takes for the amplitude of the decaying sinusoid in $c(t)$ to reach 0.02.

$$c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos(\omega_n \sqrt{1-\zeta^2}\, t - \phi) \quad \Longrightarrow \quad e^{-\zeta\omega_n t} \frac{1}{\sqrt{1-\zeta^2}} = 0.02$$

This equation is a conservative estimate, since we are assuming that $\cos(.) = 1$ at T_s. Solving for t,

$$T_s = \frac{-\ln(0.02\sqrt{1-\zeta^2})}{\zeta\omega_n}$$

The numerator of this equation varies from 3.91 to 4.74 as ζ varies from 0 to 0.9. Thus, we can **approximate** it as follows that will be used by all values of ζ.

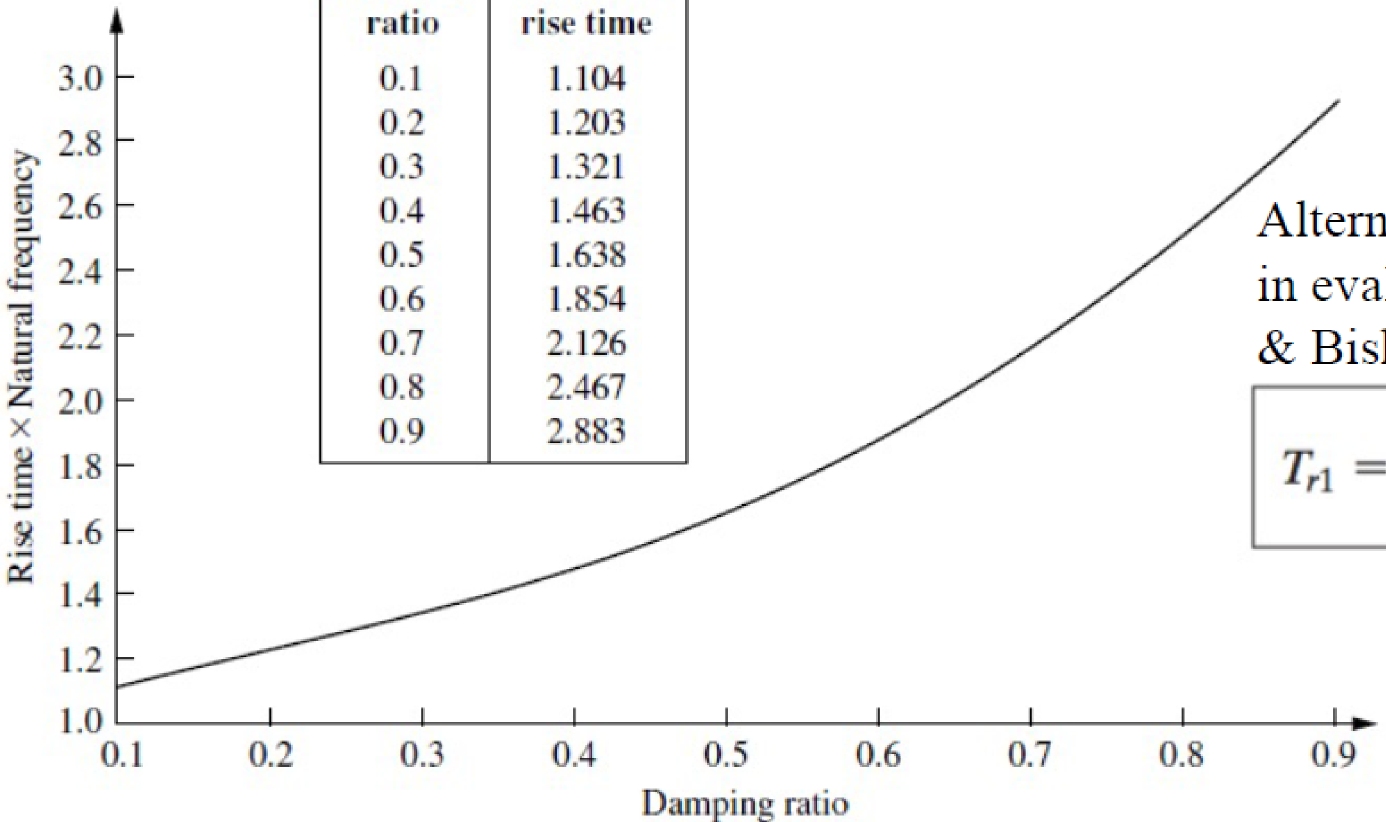
$$\boxed{T_s = \frac{4}{\zeta\omega_n}}$$

Underdamped 2nd Order systems

Unit-Step Response. The transient response specifications as function of ζ and ω_n.Rising *time*, T_r.

A precise analytical relationship between rise time and damping ratio, ζ, can not be found. However, using a computer and $c(t)$, the rise time can be found. The following figure gives a relation between natural frequency, damping ratio and rise time.

Damping ratio	Normalized rise time
0.1	1.104
0.2	1.203
0.3	1.321
0.4	1.463
0.5	1.638
0.6	1.854
0.7	2.126
0.8	2.467
0.9	2.883



Alternative approach in evaluating T_r, Dorf & Bishop:

$$\boxed{T_{r1} = \frac{2.16\zeta + 0.60}{\omega_n},}$$

Underdamped 2nd Order systems

Unit-Step Response.

Ex.

Find $T_p, T_s, \%OS$ and T_r for the transfer function given below,

$$G(s) = \frac{50}{s^2 + 15s + 100}$$

What would be final value of the unit step response?

Solution: $G(s) = \frac{50}{s^2+15s+100} = \frac{50}{s^2+2\zeta\omega_n s+\omega_n^2}$

$$\therefore\ \omega_n = \sqrt{100} = 10 \text{ rad/s and } 2\zeta\omega_n = 15 \rightarrow \zeta = \frac{15}{2\omega_n} = \frac{15}{20} = 0.75$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{3.14}{10\sqrt{1-0.75^2}} = 0.475 \text{ sec},\ T_s \cong \frac{4}{\zeta\omega_n} = \frac{4}{0.75\cdot 10} = 0.533 \text{ sec}$$

$$\%OS = 100.\,e^{-\zeta\pi/\sqrt{1-\zeta^2}} = 100.\,e^{-0.75\pi/\sqrt{1-0.75^2}} = 2.84$$

- Using the table for $\zeta = 0.75$, which corresponds to the Normalized rise time of $\omega_n T_r = 2.3 \rightarrow T_r = 0.23$ sec,
- Using the equation obtained by linearizing the nonlinear curve of normalized rise time vs damping ratio,

$$T_r = \frac{2.16\zeta+0.60}{\omega_n} = 0.222 \text{ sec}$$

Damping ratio	Normalized rise time
0.1	1.104
0.2	1.203
0.3	1.321
0.4	1.463
0.5	1.638
0.6	1.854
0.7	2.126
0.8	2.467
0.9	2.883

Underdamped 2nd Order systems

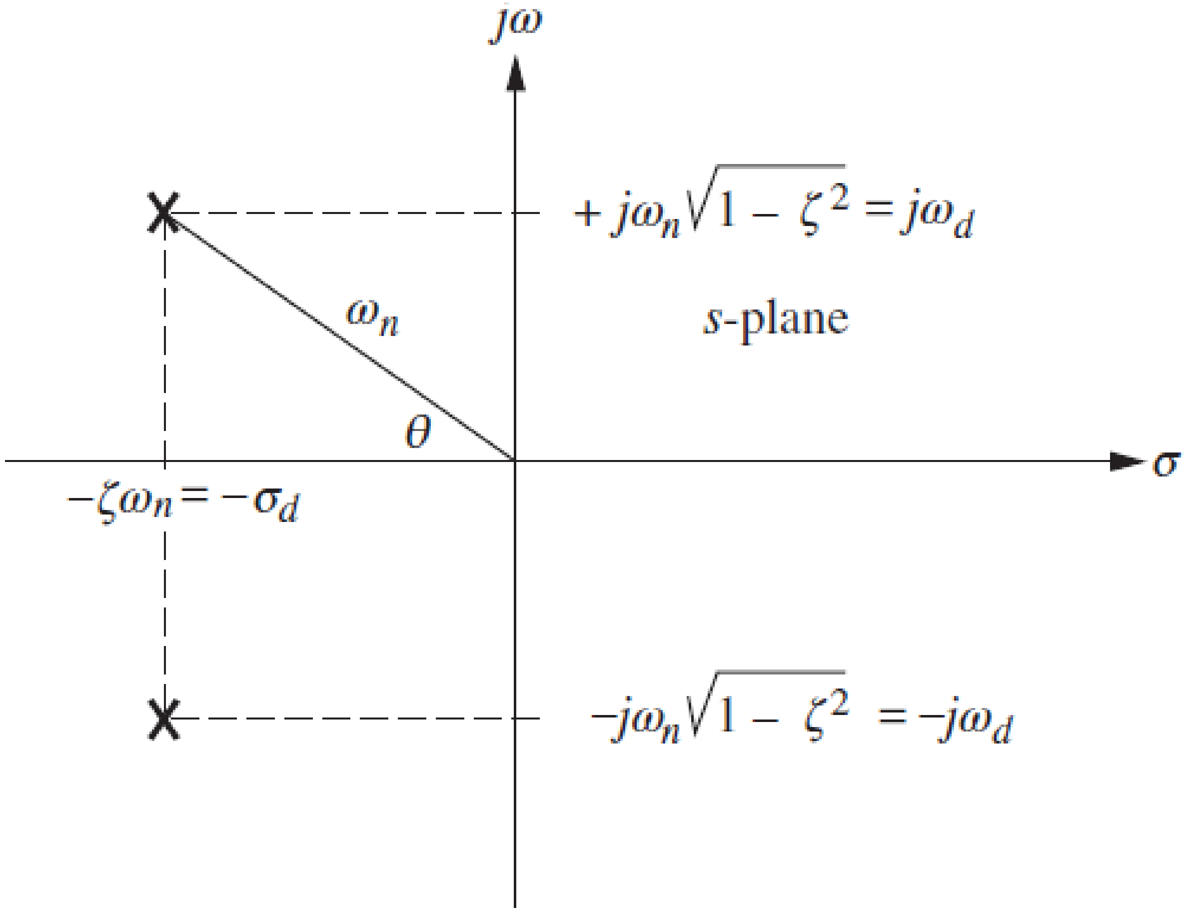
Unit-Step Response. Transient response specifications in terms of the pole location.

- From the pole plot for a general, underdamped second-order system, it is seen that the radial distance from the origin to the pole is the natural frequency, ω_n and $\cos(\theta) = \zeta$.

- Thus, *peak time* and *settling time* are:

$$\boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\mathrm{Im(poles)}} = \frac{\pi}{\omega_d}}$$

$$\boxed{T_s \cong \frac{4}{\zeta\omega_n} = \frac{4}{\mathrm{Re(poles)}} = \frac{4}{\sigma_d}}$$



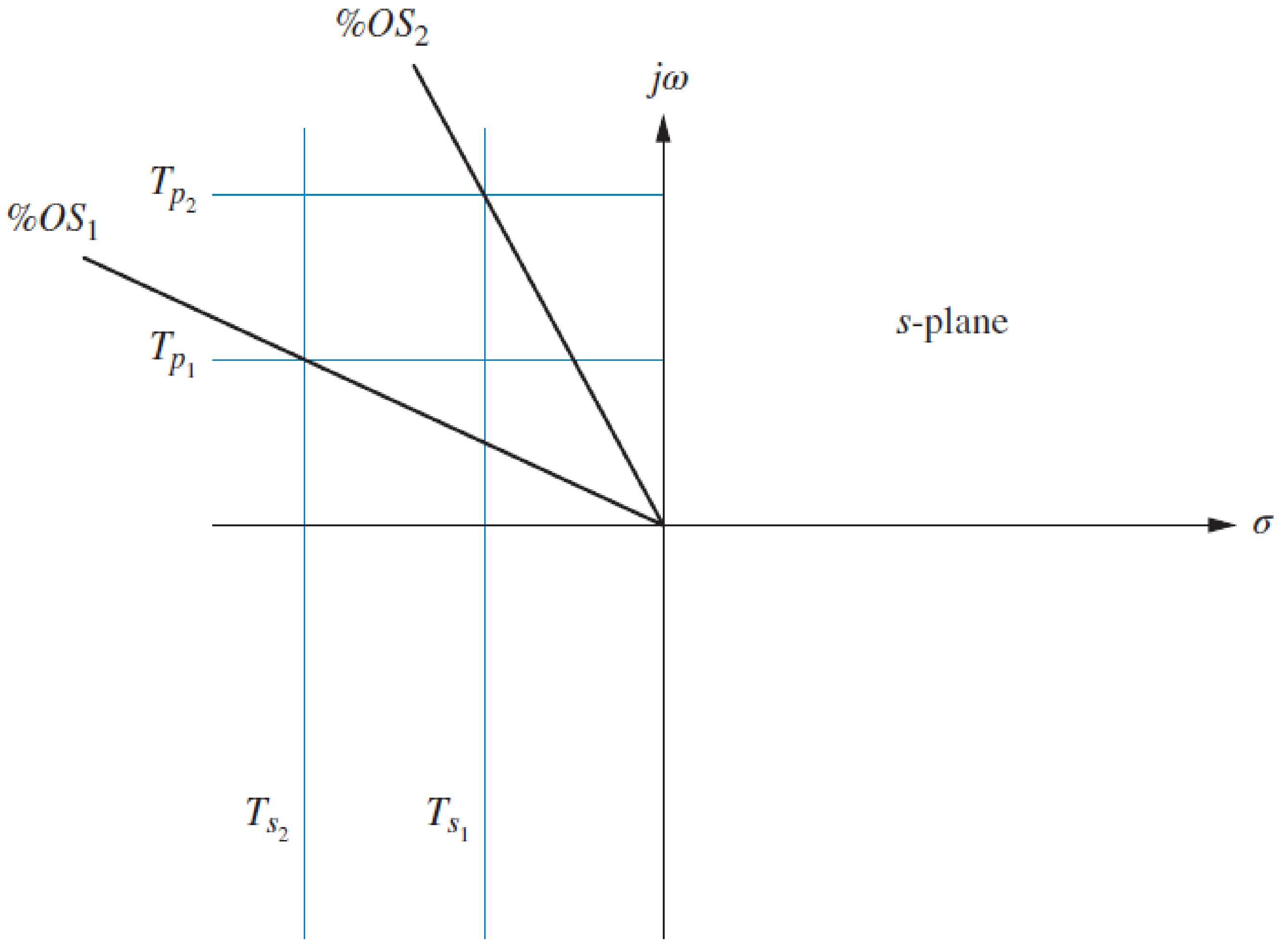
ω_d is called the *damped frequency of oscillation*, and

σ_d is the *exponential damping frequency*.

Underdamped 2nd Order systems

Unit-Step Response. Transient response specifications in terms of the pole location.

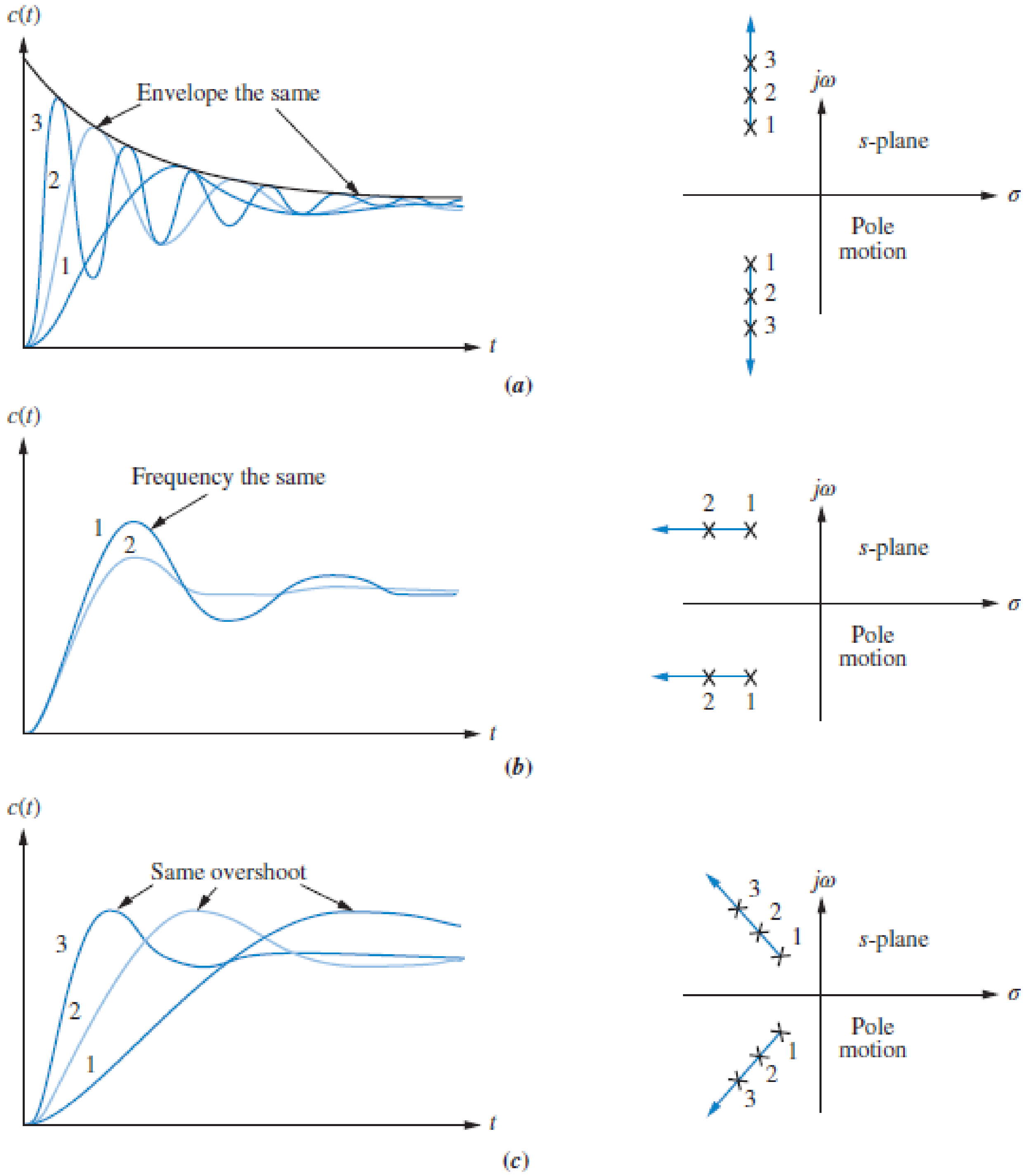
- T_p is inversely proportional to the **imaginary** part of the pole. Since **horizontal** lines on the s-plane are lines of constant imaginary value, they are also lines of **constant peak time**.
- T_s is inversely proportional to the **real** part of the pole. Since **vertical** lines on the s-plane are lines of constant real value, they are also lines of **constant settling time**.
- Since $\zeta = \cos(\theta)$, **radial** lines are lines of constant ζ. Since percent overshoot is only a function of ζ, radial lines are thus lines of **constant percent overshoot**, $\%OS$.



Underdamped 2nd Order systems

Unit-Step Response .

Transient response specifications in terms of the pole location



Underdamped 2nd Order systems

Unit-Step Response.

Ex.

Problem: Given the pole plot, find ζ, ω_n, T_p, $\%OS$ and T_s

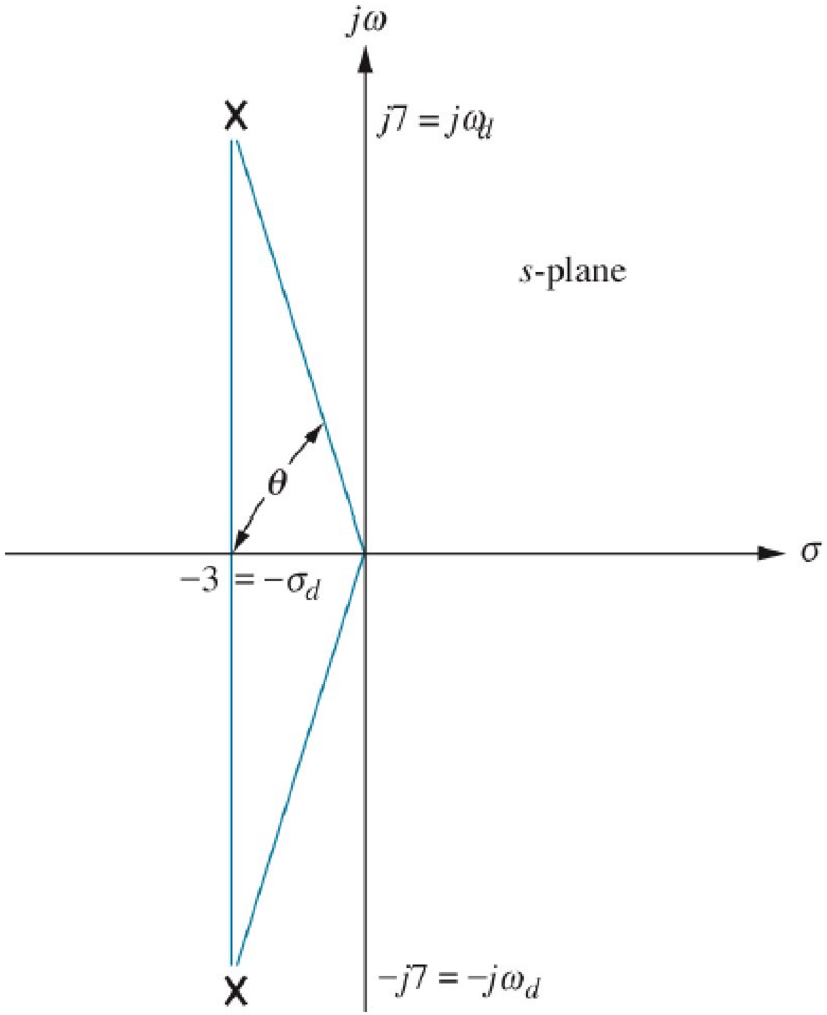


Figure 4.20

Solution:

The poles are at $-3 \pm j7$

$$\omega_n = \sqrt{3^2 + 7^2} = 7.616 \text{ rad/s}$$

$$\zeta = \cos\theta = \frac{3}{7.616} = 0.3939$$

$$\%OS = 100.\, e^{-\zeta\pi/\sqrt{1-\zeta^2}} = 26$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\text{Im(poles)}} = \frac{\pi}{7}$$

$$T_p = 0.4488 \text{ sec}$$

$$T_s \cong \frac{4}{\zeta\omega_n} = \frac{4}{\text{Re(poles)}} = \frac{4}{3}$$

$$T_s = 1.33 \text{ sec}$$

Systems with Additional Poles and Zeros

Additional Poles.

Component responses of a three-pole system: **(*a*)** pole plot; **(*b*)** component responses:

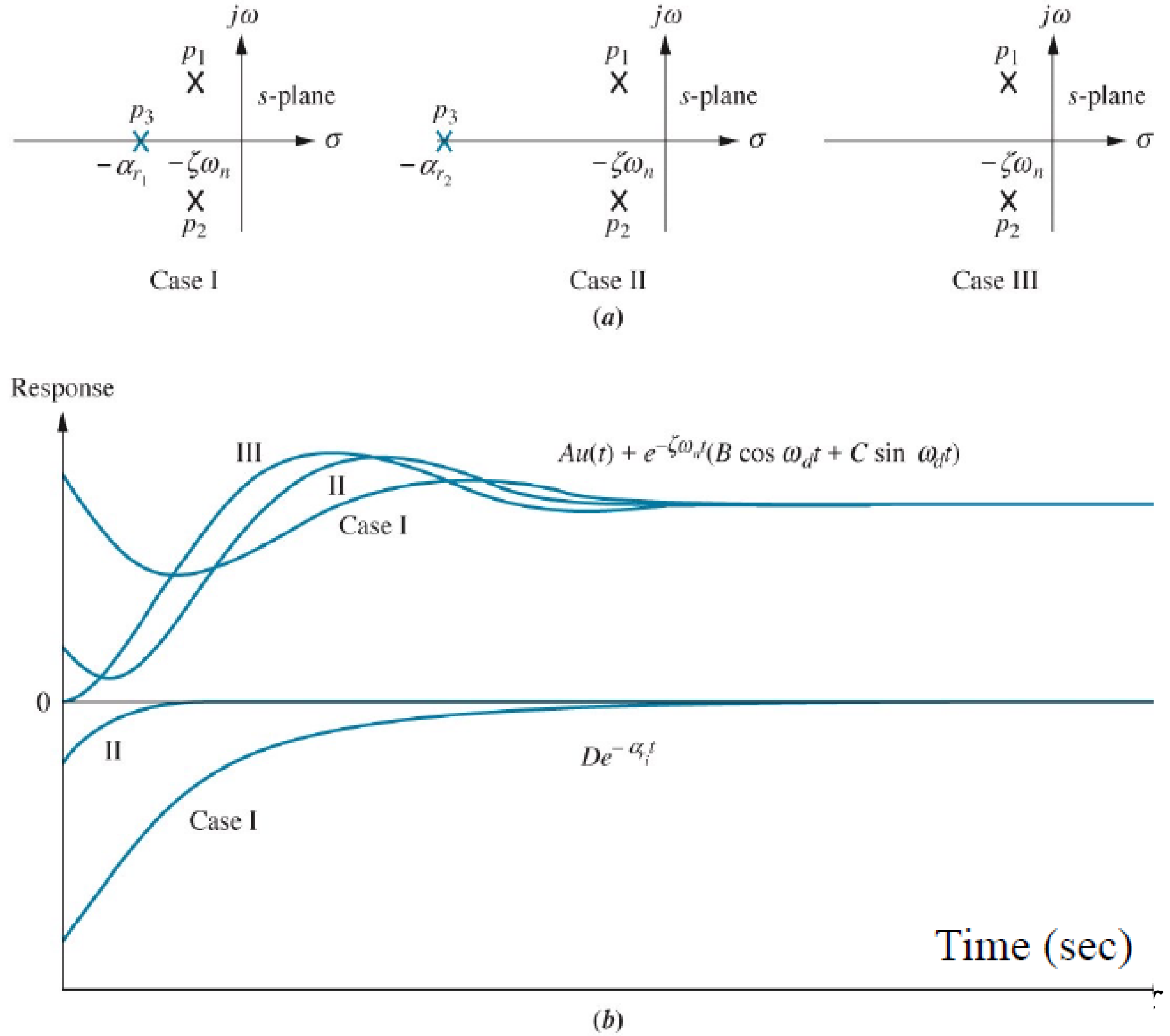
Non-dominant pole is

- near dominant second-order pair (Case I),
- far from the pair (Case II), behaves like 2nd-order systems, and
- at infinity (Case III): it is almost 2nd-order.

Conclusion:

2nd-order approximation is valid if,

$$|-\alpha_r| > 5|-\zeta\omega_n|$$



Systems with Additional Poles and Zeros

Ex.

$$T_1(s) = \frac{24.542}{s^2 + 4s + 24.542}$$

$$T_2(s) = \frac{245.42}{(s+10)(s^2 + 4s + 24.542)}$$

$$T_3(s) = \frac{73.626}{(s+3)(s^2 + 4s + 24.542)}$$

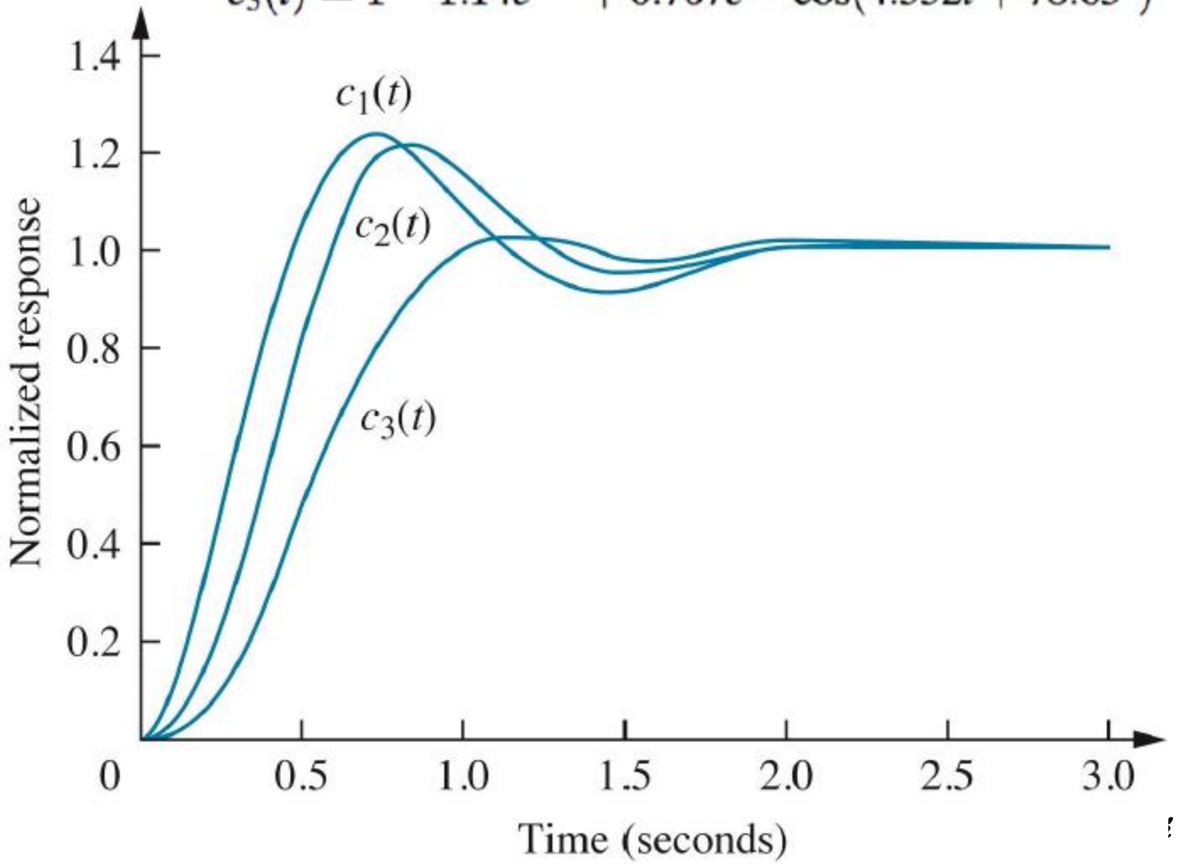
$$c_1(t) = 1 - 1.09e^{-2t}\cos(4.532t - 23.8°)$$

$$c_2(t) = 1 - 0.29e^{-10t} - 1.189e^{-2t}\cos(4.532t - 53.34°)$$

$$c_3(t) = 1 - 1.14e^{-3t} + 0.707e^{-2t}\cos(4.532t + 78.63°)$$

Step responses of system $T_1(s)$, system $T_2(s)$, and system $T_3(s)$

Conclusion: The poles are at $-2 \pm j4.53$
A pole is added first at -10 (T_2) and then at -3 (T_3).
Since -10 is much further to the left compared to -3 with respect to -2, 2nd-order approximation valid for T_2

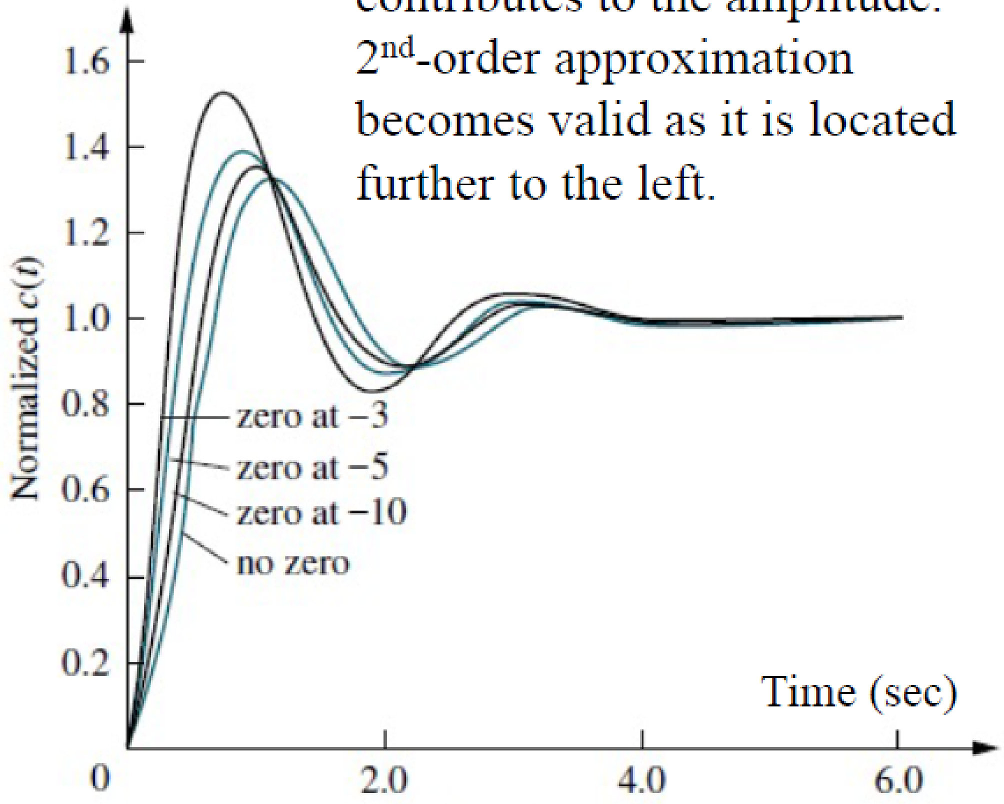


Systems with Additional Poles and Zeros

Additional Zeros.

- If a system has more than two poles or has zeros, we cannot use the formulas to calculate the performance specifications that we derived.
- However, under certain conditions, a system with more than two poles can be approximated as a second-order system that has just two complex *dominant poles*.
- We saw that the zeros of a response affect the residue, or amplitude, of a response component but do not affect the nature of the response exponential, damped sinusoid, and so on.
- Starting with a two-pole system with poles at $-1 \pm j2.828$, we consecutively add zeros at -3, -5, and -10. The results, normalized to the steady-state value, are plotted in the figure.

Conclusion: The zero contributes to the amplitude. 2^{nd}-order approximation becomes valid as it is located further to the left.



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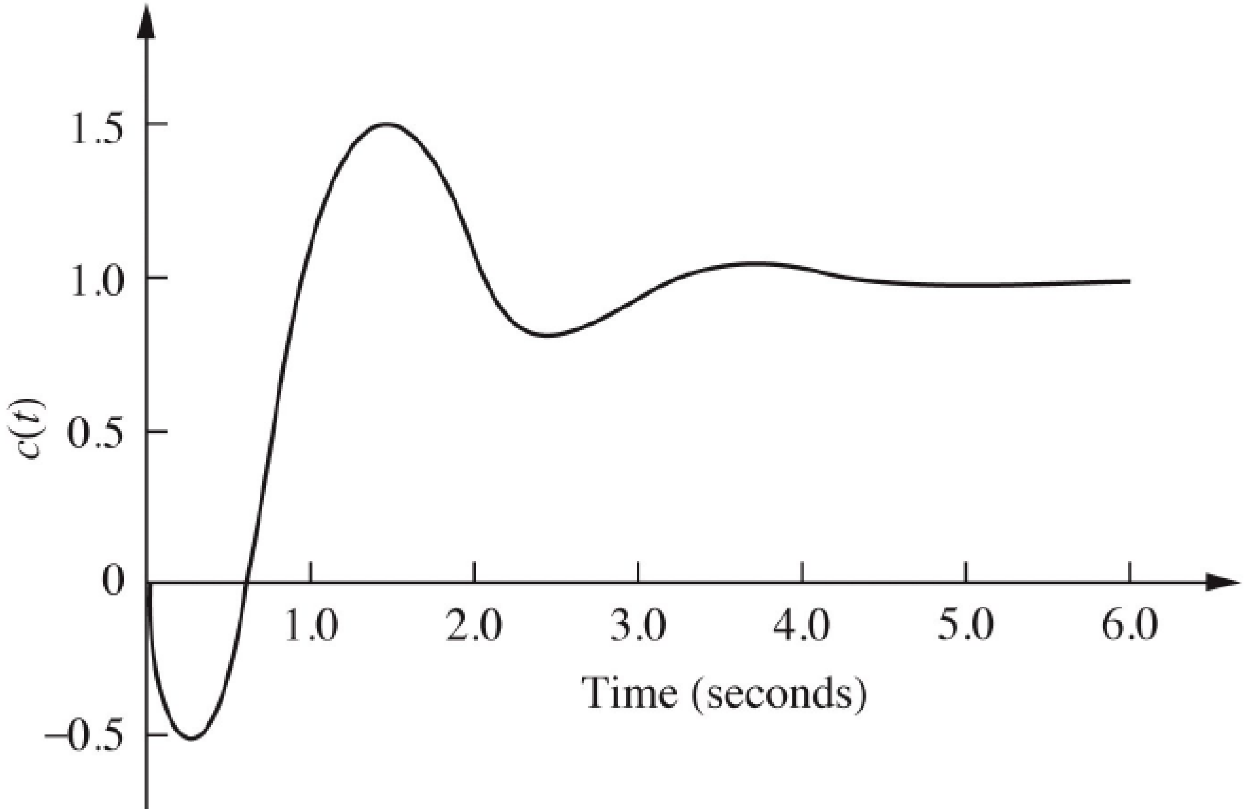
Step Response of a Nonminimum-Phase System

An interesting phenomenon occurs if there is a zero in the right half-plane (RHP).

- The response starts evolving from an opposite direction.
- This kind of systems are called nonminimum-phase systems.
- Unlike having a RHP pole, these systems having a RHP zero are not unstable but must be dealt with carefully depending on the application.



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Q6-1

- Following is the pole location for 4 different underdamped 2nd order systems. Which system has the highest peak time?

a) 3, 4

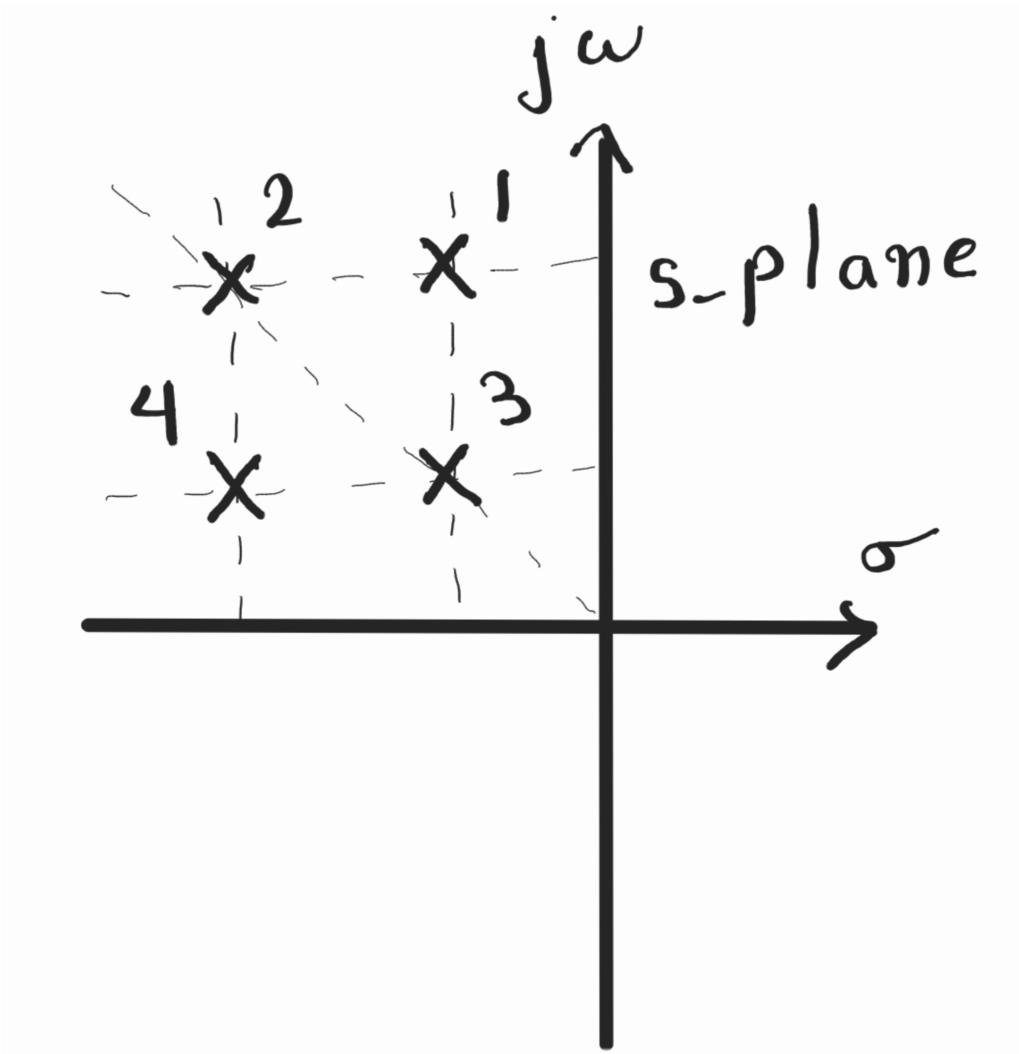
b) 1, 2

c) 1, 3

d) 2, 4

e) 2, 3

f) 1, 4



Answer: a